

# Enless Data Transmission Message Format

Version 2.7

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## 1. Frame Format on UART for wireless M Bus Message

|                 |                  |                   |              |              |              |                            |                  |                           |                          |               |               |                            |                |                |
|-----------------|------------------|-------------------|--------------|--------------|--------------|----------------------------|------------------|---------------------------|--------------------------|---------------|---------------|----------------------------|----------------|----------------|
| START<br>1 Byte | Length<br>1 Byte | C Field<br>1 Byte | M<br>2 Bytes | A<br>6 Bytes | CI<br>1 Byte | Access<br>Number<br>1 Byte | Status<br>1 Byte | Config<br>Word<br>2 Bytes | AES<br>Verify<br>2 Bytes | DIF<br>1 Byte | VIF<br>1 Byte | DATA<br>Variable<br>Length | RSSI<br>1 Byte | STOP<br>1 Byte |
|-----------------|------------------|-------------------|--------------|--------------|--------------|----------------------------|------------------|---------------------------|--------------------------|---------------|---------------|----------------------------|----------------|----------------|

### Command Structure for the RF /UART Message

- **START Byte** (0x68) – This field denotes the beginning of the frame.
- **Length Byte**- This field denotes the length of the frame excluding itself, START and STOP byte.
- **C Field** – (0x44) Fixed for the Enless products. This denotes SND-NR Command.
- **M Field** (2 Bytes) - Manufacturer ID for Enless products. It is fixed for all the products
- **A Field** (6 Bytes)

It consists of 4 Byte Unique device ID (A3-A0), device type (1 Byte) and Version (1 Byte). The 4 byte device id is packed BCD. VER and DT are related to Wireless MBus and will be ignored in the application.

Note A3 is the LO byte in the address and A0 is the HO byte, so for transmitter ID 12345678, the bytes A3 to A0 will be 78 56 34 12

|    |    |    |    |     |    |
|----|----|----|----|-----|----|
| A3 | A2 | A1 | A0 | VER | DT |
|----|----|----|----|-----|----|

- **CI** – Control Information – This field fixed to 0x7A.
- **Access Num** – Access Num of Wireless M Bus message and will be ignored.
- **Status** - Status of Wireless M Bus message and will be ignored.
- **Config Word** - Config Word of Wireless M Bus message.
  1. This will be 0000 for direct messages received by receiver
  2. This will be 0100 for repeated messages received by receiver
- **AES verify** - AES verify of Wireless M Bus message and will be ignored.
- **DIF (0x0f)** – Data Information field of Wireless M Bus message and will be ignored.
- **VIF (0x7f)** – Value Information field of Wireless M Bus message and will be ignored.
- **Data**- Variable Length data which contains Manufacturer specific commands for Enless Products.
- **RSSI** (1 Byte) – Signal strength indicator.  
Signal Strength = - RSSI (decimal) / 2 [dBm]
- **STOP Byte** – 0x16 – This denotes the end of the frame.

## 2. Periodic Data Transmission Messages

### 2.1. Temperature Sensor Data Transmit Message

| Message ID | Device Type | FirmVer | Temp Low byte | Temp High Byte | Battery Status Byte |
|------------|-------------|---------|---------------|----------------|---------------------|
| 0x01       | 0x01        | 0x01    | 0x70          | 0xFE           | 0x00                |

- **Message ID** – It will be 0x01 for the Temperature Sensor Data Transmit Message.
- **Device Type** – It will be always 0x01 for the Temperature Sensor Data Transmit Message.
- **Temp high byte and Low byte** will form a signed word for Temperature value

For Positive Temperatures, the most significant bit will be clear; hence calculation will be (Value / 10) to get actual temperature

If most significant bit is set then temperature is negative, hence calculation will be (65536 – Value) / 10 to get actual temperature

In the given example it becomes 0xFE70 which denotes the Temp -40.0°C.

- **Battery Status flag** – 0x00 for Low Battery, 0x01 for High Battery

## 2.1.1.Example Temperature Sensor Message

|               |                                                          |       |              |                |     |     |
|---------------|----------------------------------------------------------|-------|--------------|----------------|-----|-----|
| Data Frame    | 681944AE0C1000001201077A260000002F2F0F7F010101F600015716 |       |              |                |     |     |
|               | Start                                                    | Count | HEX VALUE    | DEC            |     |     |
| Start Byte    | 1                                                        | 2     | 68           |                |     |     |
| Length Byte   | 3                                                        | 2     | 19           | 25             |     |     |
| C Field       | 5                                                        | 2     | 44           |                |     |     |
| M Field       | 7                                                        | 4     | AE0C         |                |     |     |
| A Field       | 11                                                       | 12    | 100000120107 |                |     |     |
| CI Control    | 23                                                       | 2     | 7A           |                |     |     |
| Access Number | 25                                                       | 2     | 26           |                |     |     |
| Status        | 27                                                       | 2     | 00           |                |     |     |
| Config Word   | 29                                                       | 4     | 0000         |                |     |     |
| AES Verify    | 33                                                       | 4     | 2F2F         |                |     |     |
| DIF           | 37                                                       | 2     | 0F           |                |     |     |
| VIF           | 39                                                       | 2     | 7F           |                |     |     |
|               |                                                          |       |              |                | HEX | DEC |
| Data          | 41                                                       | 12    | 010101F60001 | ID             | 01  | 1   |
|               |                                                          |       |              | Type           | 01  | 1   |
|               |                                                          |       |              | Ver            | 01  | 1   |
|               |                                                          |       |              | Temp Low       | F6  | 246 |
|               |                                                          |       |              | Temp High      | 00  | 0   |
|               |                                                          |       |              | Battery Status | 01  | 1   |
|               |                                                          |       |              |                |     |     |
| RSSI          | 53                                                       | 2     | 57           |                | 57  | 87  |
| Stop          | 55                                                       | 2     | 16           |                |     |     |

## 2.2. Temperature & Humidity Sensor Data Transmit Message

| Message ID | Device Type | FirmVer | Temp Low byte | Temp High Byte | Humidity Low byte | Humidity High Byte | Battery Status Byte |
|------------|-------------|---------|---------------|----------------|-------------------|--------------------|---------------------|
| 0x02       | 0x02        | 0x01    | 0x70          | 0xFE           | 0x56              | 0x03               | 0x00                |

- **Message ID** – It will be 0x02 for the Temperature and Humidity Sensor Data Transmit Message.
- **Device Type** – It will be always 0x02 for the Temperature and Humidity Sensor Data Transmit Message.

- **Temp high byte and Low byte** will form a signed word for Temperature value

For Positive Temperatures, the most significant bit will be clear; hence calculation will be (Value / 10) to get actual temperature

If most significant bit is set then temperature is negative, hence calculation will be (65536 – Value) / 10 to get actual temperature

In the given example it becomes 0xFE70 which denotes the Temp -40.0°C.

- **Humidity Low and High Byte will form unsigned word for Humidity Value**

In the given example it becomes 0x0356 (854) which denote the Humidity 85.5 %RH

For Humidity, the most significant bit will be always clear; hence calculation will be (Value / 10) to get actual Humidity in % RH.

- **Battery Status flag** – 0x00 for Low Battery, 0x01 for High Battery

## 2.2.1.Example Temperature & Humidity Sensor Message

|               |                                                              |       |                  |                |     |     |     |       |
|---------------|--------------------------------------------------------------|-------|------------------|----------------|-----|-----|-----|-------|
| Data Frame    | 681B44AE0C1000001601077A340000002F2F0F7F020201F500A401014F16 |       |                  |                |     |     |     |       |
|               | Start                                                        | Count | HEX VALUE        | DEC            |     |     |     |       |
| Start Byte    | 1                                                            | 2     | 68               |                |     |     |     |       |
| Length Byte   | 3                                                            | 2     | 1B               | 27             |     |     |     |       |
| C Field       | 5                                                            | 2     | 44               |                |     |     |     |       |
| M Field       | 7                                                            | 4     | AE0C             |                |     |     |     |       |
| A Field       | 11                                                           | 12    | 100000160107     |                |     |     |     |       |
| CI Control    | 23                                                           | 2     | 7A               |                |     |     |     |       |
| Access Number | 25                                                           | 2     | 34               |                |     |     |     |       |
| Status        | 27                                                           | 2     | 00               |                |     |     |     |       |
| Config Word   | 29                                                           | 4     | 0000             |                |     |     |     |       |
| AES Verify    | 33                                                           | 4     | 2F2F             |                |     |     |     |       |
| DIF           | 37                                                           | 2     | 0F               |                |     |     |     |       |
| VIF           | 39                                                           | 2     | 7F               |                |     |     |     |       |
|               |                                                              |       |                  |                | HEX | DEC | DEC | Value |
| Data          | 41                                                           | 16    | 020201F500A40101 | ID             | 02  | 02  | 2   |       |
|               |                                                              |       |                  | Type           | 02  | 02  | 2   |       |
|               |                                                              |       |                  | Ver            | 01  | 01  | 1   |       |
|               |                                                              |       |                  | Temp Low       | F5  | F5  | 245 |       |
|               |                                                              |       |                  | Temo High      | 00  | 00  | 0   | 24.5  |
|               |                                                              |       |                  | Humidity Low   | A4  | A4  | 164 |       |
|               |                                                              |       |                  | Humidity High  | 01  | 01  | 1   | 42.0  |
|               |                                                              |       |                  | Battery Status | 01  | 01  | 1   |       |
| RSSI          | 57                                                           | 2     | 4F               |                | 4F  | 4F  | 79  | -39.5 |
| Stop          | 59                                                           | 2     | 16               |                |     |     |     |       |



## 2.3. PT100 Temperature Sensor Data Transmit Message

| Message ID | Device Type | FirmVer | Temp Low byte | Temp High Byte | Battery Status Byte |
|------------|-------------|---------|---------------|----------------|---------------------|
| 0x03       | 0x03        | 0x01    | 0x00          | 0x00           | 0x00                |

- **Message ID** – It will be 0x03 for the PT100 Temperature Sensor Data Transmit Message.
- **Device Type** – It will be always 0x03 for the PT100 Temperature Sensor Data Transmit Message.
- **Temp high byte and Low byte** will form a signed word for Temperature value

For Positive Temperatures, the most significant bit will be clear; hence calculation will be (Value / 10) to get actual temperature

If most significant bit is set then temperature is negative, hence calculation will be (65536 – Value) / 10 to get actual temperature

In the given example it becomes 0xFE70 which denotes the Temp -40.0°C.

- **Battery Status flag** – 0x00 for Low Battery, 0x01 for High Battery

### 2.3.1. Example PT100 Temperature Sensor Message

|               |                                                          |       |              |                |     |     |       |
|---------------|----------------------------------------------------------|-------|--------------|----------------|-----|-----|-------|
| Data Frame    | 681944AE0C2100002101077A140000002F2F0F7F0303019EFF018016 |       |              |                |     |     |       |
|               | Start                                                    | Count | HEX VALUE    | DEC            |     |     |       |
| Start Byte    | 1                                                        | 2     | 68           |                |     |     |       |
| Length Byte   | 3                                                        | 2     | 19           | 25             |     |     |       |
| C Field       | 5                                                        | 2     | 44           |                |     |     |       |
| M Field       | 7                                                        | 4     | AE0C         |                |     |     |       |
| A Field       | 11                                                       | 12    | 210000210107 |                |     |     |       |
| CI Control    | 23                                                       | 2     | 7A           |                |     |     |       |
| Access Number | 25                                                       | 2     | 14           |                |     |     |       |
| Status        | 27                                                       | 2     | 00           |                |     |     |       |
| Config Word   | 29                                                       | 4     | 0000         |                |     |     |       |
| AES Verify    | 33                                                       | 4     | 2F2F         |                |     |     |       |
| DIF           | 37                                                       | 2     | 0F           |                |     |     |       |
| VIF           | 39                                                       | 2     | 7F           |                |     |     |       |
|               |                                                          |       |              |                | HEX | DEC |       |
| Data          | 41                                                       | 12    | 0303019EFF01 | ID             | 03  | 3   |       |
|               |                                                          |       |              | Type           | 03  | 3   |       |
|               |                                                          |       |              | Ver            | 01  | 1   |       |
|               |                                                          |       |              | Temp Low       | 9E  | 158 |       |
|               |                                                          |       |              | Temp High      | FF  | 255 | -9.8  |
|               |                                                          |       |              | Battery Status | 01  | 1   |       |
|               |                                                          |       |              |                |     |     |       |
| RSSI          | 53                                                       | 2     | 80           |                | 80  | 128 | -64.0 |
| Stop          | 55                                                       | 2     | 16           |                |     |     |       |

## 2.4. Pulse Transmitter Data Transmit Message

| Message ID | Device Type | Firm Ver | Pulse Count1 D0 | Pulse Count1 D1 | Pulse Count1 D2 | Pulse Count1 D3 | Pulse Count2 D0 | Pulse Count2 D1 | Pulse Count2 D2 | Pulse Count2 D3 | Pulse Status Byte | Battery Status Byte |
|------------|-------------|----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-------------------|---------------------|
| 0x04       | 0x04        | 0x01     | 0x0A            | 0x0B            | 0x0C            | 0x0D            | 0x11            | 0x22            | 0x33            | 0xff            | 0x00              | 0x00                |

- **Message ID** – It will be 0x04 for the Pulse Transmitter Data Transmit Message.
- **Device Type** – It will be always 0x04 for the Pulse Transmitter Data Transmit Message.
- **Pulse count 1 and 2** –These are unsigned 32 bit Count which directly denotes the pulse counter reading.  
In given example Pulse Count 1 is **0x0D0C0B0A which is 218,893,066 decimal.**  
And Pulse Count 2 is **0xFF332211 which is 4,281,541,137 decimal**
- **Battery Status flag** – 0x00 for Low Battery, 0x01 for High Battery

### Pulse Status Flag:

It consists of the level information of status Inputs for Counter1 and 2 .

0 – Low level at Status Input – when contact is closed

1 – High Level at status input – when contact is open

|                    |                    |                   |                   |                    |                    |                    |                    |
|--------------------|--------------------|-------------------|-------------------|--------------------|--------------------|--------------------|--------------------|
| Counter 2 Reserved | Counter 2 Status 2 | Counter 2 Status2 | Counter 2 Status1 | Counter 1 Reserved | Counter 1 Status 3 | Counter 1 Status 2 | Counter 1 Status 1 |
| BIT7               | BIT6               | BIT5              | BIT4              | BIT3               | BIT2               | BIT1               | BIT0               |

|                            |                            |
|----------------------------|----------------------------|
| BIT4 – Counter 2 Direction | BIT0 – Counter 1 Direction |
| BIT5 – Counter 2 Tamper    | BIT1 – Counter 1 Tamper    |
| BIT6 – Counter 2 Spare     | BIT2 – Counter 1 Spare     |
| BIT7 – Counter 2 Reserved  | BIT3 – Counter 1 Reserved  |

## 2.4.1. Example Pulse Transmitter Sensor Message

|               |                                                                         |       |                             |                |     |       |
|---------------|-------------------------------------------------------------------------|-------|-----------------------------|----------------|-----|-------|
| Data Frame    | 682044AE0C1000002501077A620000002F2F0F7F0404016E00000006A0000000101AA16 |       |                             |                |     |       |
|               | Start                                                                   | Count | HEX VALUE                   | DEC            |     |       |
| Start Byte    | 1                                                                       | 2     | 68                          |                |     |       |
| Length Byte   | 3                                                                       | 2     | 20                          | 32             |     |       |
| C Field       | 5                                                                       | 2     | 44                          |                |     |       |
| M Field       | 7                                                                       | 4     | AE0C                        |                |     |       |
| A Field       | 11                                                                      | 12    | 100000250107                |                |     |       |
| CI Control    | 23                                                                      | 2     | 7A                          |                |     |       |
| Access Number | 25                                                                      | 2     | 62                          |                |     |       |
| Status        | 27                                                                      | 2     | 00                          |                |     |       |
| Config Word   | 29                                                                      | 4     | 0000                        |                |     |       |
| AES Verify    | 33                                                                      | 4     | 2F2F                        |                |     |       |
| DIF           | 37                                                                      | 2     | 0F                          |                |     |       |
| VIF           | 39                                                                      | 2     | 7F                          |                |     |       |
|               |                                                                         |       |                             |                | HEX | DEC   |
| Data          | 41                                                                      | 26    | 0404016E00000006A0000000101 | ID             | 04  | 4     |
|               |                                                                         |       |                             | Type           | 04  | 4     |
|               |                                                                         |       |                             | Ver            | 01  | 1     |
|               |                                                                         |       |                             | Pulse 1 D0     | 6E  | 110   |
|               |                                                                         |       |                             | Pulse 1 D1     | 00  | 0     |
|               |                                                                         |       |                             | Pulse 1 D2     | 00  | 0     |
|               |                                                                         |       |                             | Pulse 1 D3     | 00  | 0     |
|               |                                                                         |       |                             | Pulse 2 D0     | 6A  | 106   |
|               |                                                                         |       |                             | Pulse 2 D1     | 00  | 0     |
|               |                                                                         |       |                             | Pulse 2 D2     | 00  | 0     |
|               |                                                                         |       |                             | Pulse 2 D3     | 00  | 0     |
|               |                                                                         |       |                             | Pulse Status   | 01  | 1     |
|               |                                                                         |       |                             | Battery Status | 01  | 1     |
| RSSI          | 67                                                                      | 2     | AA                          |                | AA  | 170   |
| Stop          | 69                                                                      | 2     | 16                          |                |     | -85.0 |

## 2.5. Contact Transmitter Data Transmit Message

| Message ID | Device Type | Firm Ver | Pulse Count1 D0 | Pulse Count1 D1 | Pulse Count1 D2 | Pulse Count1 D3 | Pulse Count2 D0 | Pulse Count2 D1 | Pulse Count2 D2 | Pulse Count2 D3 | Pulse Status Byte | Battery Status Byte |
|------------|-------------|----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-------------------|---------------------|
| 0x23       | 0x23        | 0x01     | 0x0A            | 0x0B            | 0x0C            | 0x0D            | 0x11            | 0x22            | 0x33            | 0xff            | 0x00              | 0x00                |

- **Message ID** – It will be 0x23 for the Contact Transmitter Data Transmit Message.
- **Device Type** – It will be always 0x23 for the Contact Transmitter Data Transmit Message.
- **Pulse count 1 and 2** –These are unsigned 32 bit Count which directly denotes the pulse counter reading.

In given example Pulse Count 1 is **0x0D0C0B0A which is 218,893,066 decimal.**

And Pulse Count 2 is **0xFF332211 which is 4,281,541,137 decimal**

- **Battery Status flag** – 0x00 for Low Battery, 0x01 for High Battery

### Pulse/Contact Status Flag:

It consists of the level information of status Inputs for Counter1 and 2 .

0 – Low level at Status Input – when contact is closed

1 – High Level at status input – when contact is open

|                    |                    |                   |                   |                    |                    |                    |                    |
|--------------------|--------------------|-------------------|-------------------|--------------------|--------------------|--------------------|--------------------|
| Counter 2 Reserved | Counter 2 Status 2 | Counter 2 Status2 | Counter 2 Status1 | Counter 1 Reserved | Counter 1 Status 3 | Counter 1 Status 2 | Counter 1 Status 1 |
| BIT7               | BIT6               | BIT5              | BIT4              | BIT3               | BIT2               | BIT1               | BIT0               |

|                                                       |                                                       |
|-------------------------------------------------------|-------------------------------------------------------|
| BIT4 – Counter 2 Direction                            | BIT0 – Counter 1 Direction                            |
| BIT5 – Counter 2 Tamper                               | BIT1 – Counter 1 Tamper                               |
| BIT6 – Counter 2 Spare                                | BIT2 – Counter 1 Spare                                |
| BIT7 – Counter 2 Contact (0 when closed, 1 when open) | BIT3 – Counter 1 Contact (0 when closed, 1 when open) |

2.5.1.Example Contact Transmitter Sensor Message

| Data Frame                                                  |       |       |                            |                |                            |       |     |     |       | 682044AE0C1311111101077A6F0000002F2F0F7F2323010100000001000000BB015C16 |  |  |  |  |  |  |  |  |  |
|-------------------------------------------------------------|-------|-------|----------------------------|----------------|----------------------------|-------|-----|-----|-------|------------------------------------------------------------------------|--|--|--|--|--|--|--|--|--|
|                                                             | Start | Count | HEX VALUE                  | DEC            |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| Start Byte                                                  | 1     | 2     | 68                         |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| Length Byte                                                 | 3     | 2     | 20                         | 32             | count from C-Field to RSSI |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| C Field                                                     | 5     | 2     | 44                         |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| M Field                                                     | 7     | 4     | AE0C                       |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| A Field                                                     | 11    | 12    | 131111110107               |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| CI Control                                                  | 23    | 2     | 7A                         |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| Access Number                                               | 25    | 2     | 6F                         |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| Status                                                      | 27    | 2     | 00                         |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| Config Word                                                 | 29    | 4     | 0000                       |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| AES Verify                                                  | 33    | 4     | 2F2F                       |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| DIF                                                         | 37    | 2     | 0F                         |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| VIF                                                         | 39    | 2     | 7F                         |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            |                | Start                      | Count | HEX | DEC |       |                                                                        |  |  |  |  |  |  |  |  |  |
| Data                                                        | 41    | 26    | 2323010100000001000000BB01 | ID             | 41                         | 2     | 23  | 35  |       |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            | Type           | 43                         | 2     | 23  | 35  |       |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            | Ver            | 45                         | 2     | 01  | 1   |       |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            | Pulse 1 D0     | 47                         | 2     | 01  | 1   |       |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            | Pulse 1 D1     | 49                         | 2     | 00  | 0   |       |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            | Pulse 1 D2     | 51                         | 2     | 00  | 0   |       |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            | Pulse 1 D3     | 53                         | 2     | 00  | 0   | 1     |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            | Pulse 2 D0     | 55                         | 2     | 01  | 1   |       |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            | Pulse 2 D1     | 57                         | 2     | 00  | 0   |       |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            | Pulse 2 D2     | 59                         | 2     | 00  | 0   |       |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            | Pulse 2 D3     | 61                         | 2     | 00  | 0   | 1     |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            | Pulse Status   | 63                         | 2     | BB  | 187 |       |                                                                        |  |  |  |  |  |  |  |  |  |
|                                                             |       |       |                            | Battery Status | 65                         | 2     | 01  | 1   |       |                                                                        |  |  |  |  |  |  |  |  |  |
| RSSI                                                        | 67    | 2     | 5C                         |                |                            |       | 5C  | 92  | -46.0 |                                                                        |  |  |  |  |  |  |  |  |  |
| Stop                                                        | 69    | 2     | 16                         |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |
| TX CONTACT   11111113   [1011] 1   [1011] 1   -46.0dBm   OK |       |       |                            |                |                            |       |     |     |       |                                                                        |  |  |  |  |  |  |  |  |  |

## 2.6. CO2 Transmitter Data Transmit Message

- **Message ID** – It will be 0x24 for the CO2 Transmitter Data Transmit Message.

| Message ID | Device Type | Firm Ver | CO2 Low Byte D0 | CO2 High Byte D1 | Temp Low Byte D2 | Temp High Byte D3 | Hum Low Byte D4 | Hum High Byte D5 | Last min CO2 Low Byte D6 | Last min CO2 High Byte D7 | CO2 sample count | Battery Status Byte |
|------------|-------------|----------|-----------------|------------------|------------------|-------------------|-----------------|------------------|--------------------------|---------------------------|------------------|---------------------|
| 0x24       | 0x24        | 0x01     | 0x6F            | 0x01             | 0x70             | 0xFE              | 0x56            | 0x03             | 0xC2                     | 0x01                      | 0x00             | 0x00                |

- **Device Type** – It will be always 0x24 for the CO2 Transmitter Data Transmit Message.
- **CO2 high & low byte:** In the example above 0x016F = 367 therefore CO2 = 367ppm
- **Temp high byte and Low byte** will form a signed word for Temperature value

For Positive Temperatures, the most significant bit will be clear; hence calculation will be (Value / 10) to get actual temperature

If most significant bit is set then temperature is negative, hence calculation will be (65536 – Value) / 10 to get actual temperature

In the given example it becomes 0xFE70 which denotes the Temp -40.0°C.

- **Humidity Low and High Byte will form unsigned word for Humidity Value**

In the given example it becomes 0x0356 (854) which denote the Humidity 85.5 %RH

For Humidity, the most significant bit will be always clear; hence calculation will be (Value / 10) to get actual Humidity in % RH.

- **Last minimal CO2 value Low and High byte** – Reports minimal measured CO2 value since last calibration. In given example it becomes 0x01C2 (450).
- **CO2 sample count** – Reports number of CO2 samples. Samples are used for an auto calibration after defined time period.
- **Battery Status flag** – 0x00 for Low Battery, 0x01 for High Battery

2.6.1.Example CO2 Transmitter Sensor Message

|               |                                                                  |       |                      |   |                |                            |       |     |     |       |
|---------------|------------------------------------------------------------------|-------|----------------------|---|----------------|----------------------------|-------|-----|-----|-------|
| Data Frame    | 681D44AE0C0100401001077AAA0000002F2F0F7F2424016F01EA006B01016616 |       |                      |   |                |                            |       |     |     |       |
|               | Start                                                            | Count | HEX VALUE            |   | DEC            |                            |       |     |     |       |
| Start Byte    | 1                                                                | 2     | 68                   |   |                |                            |       |     |     |       |
| Length Byte   | 3                                                                | 2     | 1D                   | ➡ | 27             | count from C-Field to RSSI |       |     |     |       |
| C Field       | 5                                                                | 2     | 44                   |   |                |                            |       |     |     |       |
| M Field       | 7                                                                | 4     | AE0C                 |   |                |                            |       |     |     |       |
| A Field       | 11                                                               | 12    | 010040100107         |   |                |                            |       |     |     |       |
| CI Control    | 23                                                               | 2     | 7A                   |   |                |                            |       |     |     |       |
| Access Number | 25                                                               | 2     | AA                   |   |                |                            |       |     |     |       |
| Status        | 27                                                               | 2     | 00                   |   |                |                            |       |     |     |       |
| Config Word   | 29                                                               | 4     | 0000                 |   |                |                            |       |     |     |       |
| AES Verify    | 33                                                               | 4     | 2F2F                 |   |                |                            |       |     |     |       |
| DIF           | 37                                                               | 2     | 0F                   |   |                |                            |       |     |     |       |
| VIF           | 39                                                               | 2     | 7F                   |   |                |                            |       |     |     |       |
|               |                                                                  |       |                      |   |                | Start                      | Count | HEX | DEC | DEC   |
| Data          | 41                                                               | 20    | 2424016F01EA006B0101 |   | ID             | 41                         | 2     | 24  | 24  | 36    |
|               |                                                                  |       |                      |   | Type           | 43                         | 2     | 24  | 24  | 36    |
|               |                                                                  |       |                      |   | Ver            | 45                         | 2     | 01  | 01  | 1     |
|               |                                                                  |       |                      |   | CO2 Low        | 47                         | 2     | 6F  | 6F  | 111   |
|               |                                                                  |       |                      |   | CO2 High       | 49                         | 2     | 01  | 01  | 1     |
|               |                                                                  |       |                      |   | Temp Low       | 51                         | 2     | EA  | EA  | 234   |
|               |                                                                  |       |                      |   | Temo High      | 53                         | 2     | 00  | 00  | 0     |
|               |                                                                  |       |                      |   | Humidity Low   | 55                         | 2     | 6B  | 6B  | 107   |
|               |                                                                  |       |                      |   | Humidity High  | 57                         | 2     | 01  | 01  | 1     |
|               |                                                                  |       |                      |   | Battery Status | 59                         | 2     | 01  | 01  | 1     |
|               |                                                                  |       |                      |   |                |                            |       |     |     |       |
| RSSI          | 61                                                               | 2     | 66                   |   |                |                            |       | 66  | 66  | 102   |
| Stop          | 63                                                               | 2     | 16                   |   |                |                            |       |     |     | -51.0 |

TX CO2 | 10400001 | 367ppm | 23.4°C 36.3%RH | -51.0dBm | OK



## 2.7. 4-20mA Analogue Transmitter Data Transmit Message

| Message ID | Device Type | Firm Ver | Value Low D0 | Value High D1 | Battery Status Byte |
|------------|-------------|----------|--------------|---------------|---------------------|
| 0x25       | 0x25        | 0x01     | 0x68         | 0x01          | 0x00                |

- **Message ID** – It will be 0x25 for the 4-20mA Analogue Transmitter Data Transmit Message.
- **Device Type** – It will be always 0x25 for the 4-20mA Analogue Transmitter Data Transmit Message.
- **Value High & Low Bytes** – In the example shown above  $0x0168 = 360$ . Divide this by 100 to get 3.60mA
- **Battery Status flag** – 0x00 for Low Battery, 0x01 for High Battery

2.7.1.Example 4-20mA Analogue Transmitter Sensor Message

| Data Frame                                      |       | 681944AE0C1011111101077A6B0000002F2F0F7F2626016801016C16 |              |                |                            |       |     |     |       |  |  |
|-------------------------------------------------|-------|----------------------------------------------------------|--------------|----------------|----------------------------|-------|-----|-----|-------|--|--|
|                                                 | Start | Count                                                    | HEX VALUE    | DEC            |                            |       |     |     |       |  |  |
| Start Byte                                      | 1     | 2                                                        | 68           |                |                            |       |     |     |       |  |  |
| Length Byte                                     | 3     | 2                                                        | 19           | 25             | count from C-Field to RSSI |       |     |     |       |  |  |
| C Field                                         | 5     | 2                                                        | 44           |                |                            |       |     |     |       |  |  |
| M Field                                         | 7     | 4                                                        | AE0C         |                |                            |       |     |     |       |  |  |
| A Field                                         | 11    | 12                                                       | 101111110107 |                |                            |       |     |     |       |  |  |
| CI Control                                      | 23    | 2                                                        | 7A           |                |                            |       |     |     |       |  |  |
| Access Number                                   | 25    | 2                                                        | 6B           |                |                            |       |     |     |       |  |  |
| Status                                          | 27    | 2                                                        | 00           |                |                            |       |     |     |       |  |  |
| Config Word                                     | 29    | 4                                                        | 0000         |                |                            |       |     |     |       |  |  |
| AES Verify                                      | 33    | 4                                                        | 2F2F         |                |                            |       |     |     |       |  |  |
| DIF                                             | 37    | 2                                                        | 0F           |                |                            |       |     |     |       |  |  |
| VIF                                             | 39    | 2                                                        | 7F           |                |                            |       |     |     |       |  |  |
| Data                                            | 41    | 12                                                       | 262601680101 | ID             | Start                      | Count | HEX | DEC |       |  |  |
|                                                 |       |                                                          |              | Type           | 41                         | 2     | 26  | 38  |       |  |  |
|                                                 |       |                                                          |              | Ver            | 43                         | 2     | 26  | 38  |       |  |  |
|                                                 |       |                                                          |              | Value Low      | 45                         | 2     | 01  | 1   |       |  |  |
|                                                 |       |                                                          |              | Value High     | 47                         | 2     | 68  | 104 |       |  |  |
|                                                 |       |                                                          |              | Battery Status | 49                         | 2     | 01  | 1   | 360   |  |  |
|                                                 |       |                                                          |              | 51             | 2                          | 01    | 1   |     |       |  |  |
| RSSI                                            | 53    | 2                                                        | 6C           |                |                            |       | 6C  | 108 | -54.0 |  |  |
| Stop                                            | 55    | 2                                                        | 16           |                |                            |       |     |     |       |  |  |
| TX 4-20mA   11111112   3.60mA     -37.5dBm   OK |       |                                                          |              |                |                            |       |     |     |       |  |  |

## 2.8. 0-5V Analogue Transmitter Data Transmit Message

| Message ID | Device Type | Firm Ver | Value Low D0 | Value High D1 | Battery Status Byte |
|------------|-------------|----------|--------------|---------------|---------------------|
| 0x26       | 0x26        | 0x01     | 0x67         | 0x01          | 0x00                |

- **Message ID** – It will be 0x26 for the 0-5V Analogue Transmitter Data Transmit Message.
- **Device Type** – It will be always 0x26 for the 0-5V Analogue Transmitter Data Transmit Message.
- **Value High & Low Bytes** – In the example shown above  $0x0167 = 359$ . Divide this by 100 to get 3.59V
- **Battery Status flag** – 0x00 for Low Battery, 0x01 for High Battery

2.8.1.Example 0-5V Analogue Transmitter Sensor Message

| Data Frame                                   |       |       |              |                |                            |       |     |     |       | 681944AE0C101111101077A6B0000002F2F0F7F2626016701016C16 |  |  |  |  |  |  |  |  |  |
|----------------------------------------------|-------|-------|--------------|----------------|----------------------------|-------|-----|-----|-------|---------------------------------------------------------|--|--|--|--|--|--|--|--|--|
|                                              | Start | Count | HEX VALUE    | DEC            |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| Start Byte                                   | 1     | 2     | 68           |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| Length Byte                                  | 3     | 2     | 19           | 25             | count from C-Field to RSSI |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| C Field                                      | 5     | 2     | 44           |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| M Field                                      | 7     | 4     | AE0C         |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| A Field                                      | 11    | 12    | 101111110107 |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| CI Control                                   | 23    | 2     | 7A           |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| Access Number                                | 25    | 2     | 6B           |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| Status                                       | 27    | 2     | 00           |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| Config Word                                  | 29    | 4     | 0000         |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| AES Verify                                   | 33    | 4     | 2F2F         |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| DIF                                          | 37    | 2     | 0F           |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| VIF                                          | 39    | 2     | 7F           |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
|                                              |       |       |              |                | Start                      | Count | HEX | DEC |       |                                                         |  |  |  |  |  |  |  |  |  |
| Data                                         | 41    | 12    | 262601670101 | ID             | 41                         | 2     | 26  | 38  |       |                                                         |  |  |  |  |  |  |  |  |  |
|                                              |       |       |              | Type           | 43                         | 2     | 26  | 38  |       |                                                         |  |  |  |  |  |  |  |  |  |
|                                              |       |       |              | Ver            | 45                         | 2     | 01  | 1   |       |                                                         |  |  |  |  |  |  |  |  |  |
|                                              |       |       |              | Value Low      | 47                         | 2     | 67  | 103 |       |                                                         |  |  |  |  |  |  |  |  |  |
|                                              |       |       |              | Value High     | 49                         | 2     | 01  | 1   | 359   |                                                         |  |  |  |  |  |  |  |  |  |
|                                              |       |       |              | Battery Status | 51                         | 2     | 01  | 1   |       |                                                         |  |  |  |  |  |  |  |  |  |
| RSSI                                         | 53    | 2     | 6C           |                |                            |       | 6C  | 108 | -54.0 |                                                         |  |  |  |  |  |  |  |  |  |
| Stop                                         | 55    | 2     | 16           |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |
| TX 0-5V   11111110   3.59V     -54.0dBm   OK |       |       |              |                |                            |       |     |     |       |                                                         |  |  |  |  |  |  |  |  |  |

## 2.9. 0-10V Analogue Transmitter Data Transmit Message

| Message ID | Device Type | Firm Ver | Value Low D0 | Value High D1 | Battery Status Byte |
|------------|-------------|----------|--------------|---------------|---------------------|
| 0x27       | 0x27        | 0x01     | 0x67         | 0x01          | 0x00                |

- **Message ID** – It will be 0x27 for the 0-10V Analogue Transmitter Data Transmit Message.
- **Device Type** – It will be always 0x27 for the 0-10V Analogue Transmitter Data Transmit Message.
- **Value High & Low Bytes** – In the example shown above  $0x0167 = 359$ . Divide this by 100 to get 3.59V
- **Battery Status flag** – 0x00 for Low Battery, 0x01 for High Battery

2.9.1.Example 0-10V Analogue Transmitter Sensor Message

|                                               |                                                         |       |              |                |                            |       |     |     |       |     |
|-----------------------------------------------|---------------------------------------------------------|-------|--------------|----------------|----------------------------|-------|-----|-----|-------|-----|
| Data Frame                                    | 681944AE0C111111101077A700000002F2F0F7F2727016701015816 |       |              |                |                            |       |     |     |       |     |
|                                               | Start                                                   | Count | HEX VALUE    | DEC            |                            |       |     |     |       |     |
| Start Byte                                    | 1                                                       | 2     | 68           | 25             | count from C-Field to RSSI |       |     |     |       |     |
| Length Byte                                   | 3                                                       | 2     | 19           |                |                            |       |     |     |       |     |
| C Field                                       | 5                                                       | 2     | 44           |                |                            |       |     |     |       |     |
| M Field                                       | 7                                                       | 4     | AE0C         |                |                            |       |     |     |       |     |
| A Field                                       | 11                                                      | 12    | 111111110107 |                |                            |       |     |     |       |     |
| CI Control                                    | 23                                                      | 2     | 7A           |                |                            |       |     |     |       |     |
| Access Number                                 | 25                                                      | 2     | 70           |                |                            |       |     |     |       |     |
| Status                                        | 27                                                      | 2     | 00           |                |                            |       |     |     |       |     |
| Config Word                                   | 29                                                      | 4     | 0000         |                |                            |       |     |     |       |     |
| AES Verify                                    | 33                                                      | 4     | 2F2F         |                |                            |       |     |     |       |     |
| DIF                                           | 37                                                      | 2     | 0F           |                |                            |       |     |     |       |     |
| VIF                                           | 39                                                      | 2     | 7F           |                |                            |       |     |     |       |     |
| Data                                          | 41                                                      | 12    | 272701670101 | ID             | Start                      | Count | HEX | DEC |       |     |
|                                               |                                                         |       |              | Type           | 41                         | 2     | 27  | 39  |       |     |
|                                               |                                                         |       |              | Ver            | 43                         | 2     | 27  | 39  |       |     |
|                                               |                                                         |       |              | Value Low      | 45                         | 2     | 01  | 1   |       |     |
|                                               |                                                         |       |              | Value High     | 47                         | 2     | 67  | 103 |       |     |
|                                               |                                                         |       |              | Value High     | 49                         | 2     | 01  | 1   |       | 359 |
|                                               |                                                         |       |              | Battery Status | 51                         | 2     | 01  | 1   |       |     |
| RSSI                                          | 53                                                      | 2     | 58           |                |                            |       | 58  | 88  | -44.0 |     |
| Stop                                          | 55                                                      | 2     | 16           |                |                            |       |     |     |       |     |
| TX 0-10V   11111111   3.59V     -44.0dBm   OK |                                                         |       |              |                |                            |       |     |     |       |     |

## 2.10. Low battery Message

| Message ID | Device Type | FirmVer | Battery Level |
|------------|-------------|---------|---------------|
| 0x05       | 0x04        | 0x01    | 0x00          |

**This message is NOT used in the Enless Products**

## 2.11. Energy Meter Data Transmit Messages

| Message ID | Device Type | Firm Ver | Pulse Count1 D0 | Pulse Count1 D1 | Pulse Count1 D2 | Pulse Count1 D3 | Pulse Count2 D0 | Pulse Count2 D1 | Pulse Count2 D2 | Pulse Count2 D3 | Pulse Status Byte |
|------------|-------------|----------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-------------------|
| 0x06       | 0x05        | 0x01     | 0x0A            | 0x0B            | 0x0C            | 0x0D            | 0x11            | 0x22            | 0x33            | 0xff            | 0x00              |

- **Message ID** – It will be 0x06 for the Energy Meter Transmitter Data Transmit Message.
- **Device Type** – It will be always 0x05 for the Energy Meter Transmitter Data Transmit Message.
- **Pulse count 1 and 2** –These are unsigned 32 bit Count which directly denotes the pulse counter reading.  
In given example Pulse Count 1 is **0x0D0C0B0A which is 218,893,066 decimal.**  
And Pulse Count 2 is **0xFF332211 which is 4,281,541,137 decimal**
- **Pulse Status Flag:**

It consists of the level information of status Inputs for Counter1 and 2 .

0 – Low level at Status Input

1 – High Level at status input

|                    |                    |                   |                   |                    |                    |                    |                    |
|--------------------|--------------------|-------------------|-------------------|--------------------|--------------------|--------------------|--------------------|
| Counter 2 Reserved | Counter 2 Status 2 | Counter 2 Status2 | Counter 2 Status1 | Counter 1 Reserved | Counter 1 Status 3 | Counter 1 Status 2 | Counter 1 Status 1 |
| BIT7               | BIT6               | BIT5              | BIT4              | BIT3               | BIT2               | BIT1               | BIT0               |



### 2.11.1. *Energy Meter Transmitter - Modbus Slave- Data transmit Message.*

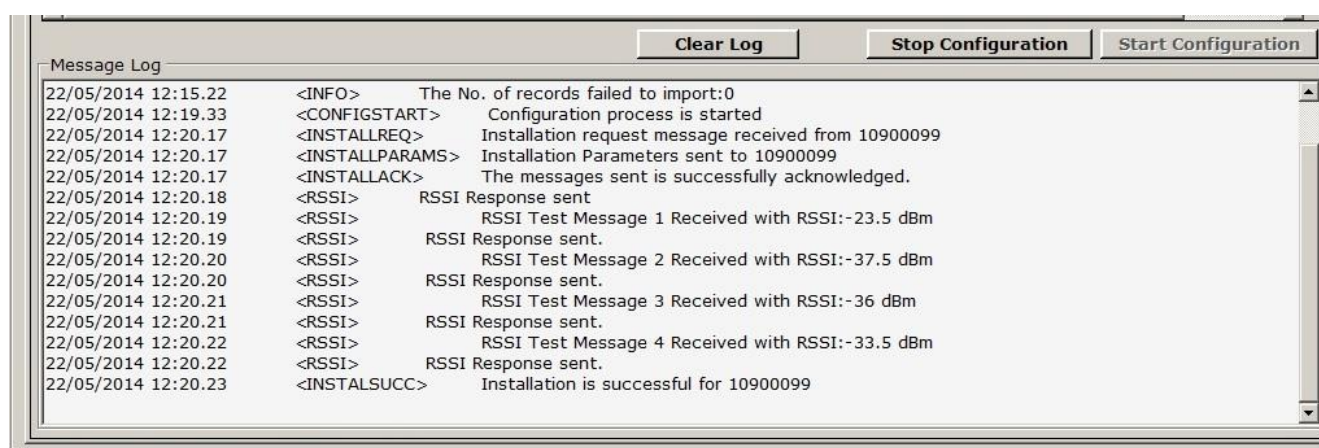
| Msg ID | Device Type | Firm Ver | Message Num | Slave Id | ModBus Meter data D1 | ModBus Meter data D0 | ModBus Meter data D3 | ModBus Meter data D2 | ModBus Meter data D5 | ModBus Meter data D4 | ModBus Meter data D7 | ModBus Meter data D6 | ModBus Meter data D9 | ModBus Meter data D8 |
|--------|-------------|----------|-------------|----------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| 0x07   | 0x05        | 0x01     | 0x01 – 0x0A | 0x01     | 0x22                 | 0x11                 | 0x44                 | 0x33                 | 0x66                 | 0x55                 | 0x88                 | 0x77                 | 0xAA                 | 0x99                 |

- **Message ID** – It will be 0x07 for the Energy Meter Transmitter Modbus slave Data Transmit Message.
- **Device Type** – It will be always 0x05 for the Energy Meter Transmitter Data Transmit Message.
- **Message Num** – This gives the message number for the 1-10 slaves of the Meter. Message Num can be 1 -10.
- **Slave Id** – Modbus Slave ID 0x00-0xFF
- **Modbus Meter Data (D0-D9)** - This is a raw data read from the 5 Modbus Registers configured at the installation Mode. This is a raw data and the conversion / meaning can be known by seeing the Register specification at Modbus Slaves (Meters).  
D1D0 – Register 1 = 0x2211  
D3D2 – Register 2 =0x4433  
D5D4– Register 3 = 0x6655  
D7D6 – Register 4= 0x8877  
D9D8 – Register 5 = 0xAA99

### 3. Field Installation Messages (Over-The-Air)

For Field installation, the process is initiated by the transmitters. The host replies back to the messages which includes the configuration data and acknowledgements to messages. This section details the messages as they would be visible 'over-the-air' and typically used by manufacturers that use their own wireless modules.

Below is a typical sequence seen on the field installation tool for all transmitters, except the Energy Meter Transmitter:



The sequence of communications is as follows:

| No. | Time     | Direction | Description                            | Message Type                  |
|-----|----------|-----------|----------------------------------------|-------------------------------|
| 1   | 12:20:17 | Tx→Host   | Installation Request Received from Tx. | Installation Request (0x10)   |
| 2   | 12:20:17 | Host→Tx   | Installation Parameters Sent to Tx.    | Installation Parameter (0x11) |
| 3   | 12:20:17 | Tx→Host   | Acknowledgement Received From Tx.      | Installation ACK (0x12)       |
| 4   | 12:20:18 | Host→Tx   | RSSI response Sent to Tx.              | RSSI Response (0x15)          |
| 5   | 12:20:19 | Tx→Host   | RSSI 1 Message Received From Tx.       | Installation RSSI (0x14)      |
| 6   | 12:20:19 | Host→Tx   | RSSI 1 response Sent to Tx.            | RSSI Response (0x15)          |
| 7   | 12:20:20 | Tx→Host   | RSSI 2 Message Received From Tx.       | Installation RSSI (0x14)      |
| 8   | 12:20:20 | Host→Tx   | RSSI 2 response Sent to Tx.            | RSSI Response (0x15)          |
| 9   | 12:20:21 | Tx→Host   | RSSI 3 Message Received From Tx.       | Installation RSSI (0x14)      |
| 10  | 12:20:21 | Host→Tx   | RSSI 3 response Sent to Tx.            | RSSI Response (0x15)          |
| 11  | 12:20:22 | Tx→Host   | RSSI 4 Message Received From Tx.       | Installation RSSI (0x14)      |
| 12  | 12:20:22 | Host→Tx   | RSSI 4 response Sent to Tx.            | RSSI Response (0x15)          |

## Enless Data Transmission Message Format

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|    |          |         |                         |                                |
|----|----------|---------|-------------------------|--------------------------------|
| 13 | 12:20:23 | Tx→Host | Installation Successful | Installation Successful (0x16) |
| 14 |          | Tx→Host | Installation Successful | Installation Successful (0x16) |

Some Complete Installation Sequence Packets (over-the-air) can be seen below (with the packet types highlighted in Red:

Pulse transmitter:

```
09:44.10 681544AE0C9900901001077A010000002F2F0F7F10043516
09:44.10 68284424487856341201077A0A0000002F2F0F7F110401109000990500010040200000000A000000643516
09:44.11 681644AE0C9900901001077A020000002F2F0F7F1204003C16
09:44.12 681B4424487856341201077A0A0000002F2F0F7F15040110900099533316
09:44.12 681644AE0C9900901001077A030000002F2F0F7F1404024A16
09:44.13 681B4424487856341201077A0C0000002F2F0F7F150402109000993F3216
09:44.13 681644AE0C9900901001077A040000002F2F0F7F1404034816
09:44.14 681B4424487856341201077A0D0000002F2F0F7F15040310900099403216
09:44.14 681644AE0C9900901001077A050000002F2F0F7F1404044616
09:44.15 681B4424487856341201077A0E0000002F2F0F7F15040410900099423216
09:44.15 681644AE0C9900901001077A060000002F2F0F7F1404054316
09:44.16 681B4424487856341201077A0F0000002F2F0F7F15040510900099423216
09:44.16 681544AE0C9900901001077A070000002F2F0F7F16044416
09:44.17 681544AE0C9900901001077A080000002F2F0F7F16044316
```

Temperature and Humidity Transmitter:

```
09:44.36 681544AE0C6100201001077A010000002F2F0F7F10023A16
09:44.37 68204424487856341201077A140000002F2F0F7F110201102000610500010040203216
09:44.37 681644AE0C6100201001077A020000002F2F0F7F1202003B16
09:44.39 681B4424487856341201077A140000002F2F0F7F150201102000612F3216
09:44.39 681644AE0C6100201001077A030000002F2F0F7F1402023816
09:44.39 681B4424487856341201077A160000002F2F0F7F150202102000612D3216
09:44.40 681644AE0C6100201001077A040000002F2F0F7F1402033716
09:44.40 681B4424487856341201077A170000002F2F0F7F150203102000612D3116
09:44.41 681644AE0C6100201001077A050000002F2F0F7F1402043816
09:44.41 681B4424487856341201077A180000002F2F0F7F150204102000612C3216
09:44.42 681644AE0C6100201001077A060000002F2F0F7F1402053816
09:44.42 681B4424487856341201077A190000002F2F0F7F150205102000612C3216
09:44.43 681544AE0C6100201001077A070000002F2F0F7F16023716
09:44.43 681544AE0C6100201001077A080000002F2F0F7F16023716
```

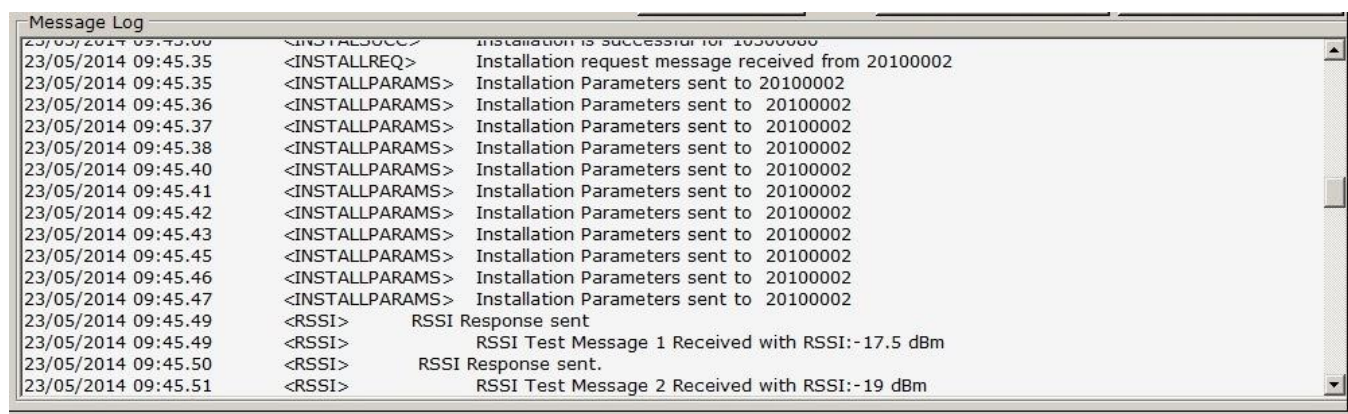
PT100 Transmitter:

```
09:45.02 681544AE0C8600501001077A010000002F2F0F7F10033316
09:45.03 68204424487856341201077A1E0000002F2F0F7F110301105000860500010040204616
09:45.03 681644AE0C8600501001077A020000002F2F0F7F1203003416
09:45.04 681B4424487856341201077A1E0000002F2F0F7F15030110500086333216
09:45.05 681644AE0C8600501001077A030000002F2F0F7F1403023516
09:45.05 681B4424487856341201077A200000002F2F0F7F15030210500086323016
09:45.06 681644AE0C8600501001077A040000002F2F0F7F1403033416
09:45.06 681B4424487856341201077A210000002F2F0F7F15030310500086323016
09:45.07 681644AE0C8600501001077A050000002F2F0F7F1403043416
09:45.07 681B4424487856341201077A220000002F2F0F7F15030410500086323216
09:45.08 681644AE0C8600501001077A060000002F2F0F7F1403053416
```

## Enless Data Transmission Message Format

09:45:08 681B4424487856341201077A230000002F2F0F7F15030510500086323616  
09:45:09 681544AE0C8600501001077A070000002F2F0F7F16033516  
09:45:09 681544AE0C8600501001077A080000002F2F0F7F16033516

For the Energy Meter Transmitter, below is the sequence:



| Message Log         |                 |                                                     |
|---------------------|-----------------|-----------------------------------------------------|
| 23/05/2014 09:45:08 | <INSTALLREQ>    | Installation is successful for 10000000             |
| 23/05/2014 09:45:35 | <INSTALLREQ>    | Installation request message received from 20100002 |
| 23/05/2014 09:45:35 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:36 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:37 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:38 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:40 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:41 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:42 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:43 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:45 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:46 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:47 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:49 | <RSSI>          | RSSI Response sent                                  |
| 23/05/2014 09:45:49 | <RSSI>          | RSSI Test Message 1 Received with RSSI:-17.5 dBm    |
| 23/05/2014 09:45:50 | <RSSI>          | RSSI Response sent.                                 |
| 23/05/2014 09:45:51 | <RSSI>          | RSSI Test Message 2 Received with RSSI:-19 dBm      |

The sequence of communications for EMT Installation is as follows:

| No. | Time     | Direction | Description                            | Message Type                        |
|-----|----------|-----------|----------------------------------------|-------------------------------------|
| 1   | 09:45:35 | Tx→Host   | Installation Request Received from Tx. | Parameter Request (0x10)            |
| 2   | 09:45:35 | Host→Tx   | Installation Parameters Sent to Tx.    | Field Installation Parameter (0x11) |
| 3   |          | Tx→Host   | Request Parameters Slave 1             | Modbus Parameter Request (0x12)     |
| 4   | 09:45:36 | Host→Tx   | Parameters for Modbus Slave 1          | Modbus Parameter (0x11)             |
| 5   |          | Tx→Host   | Request Parameters Slave 2             | Modbus Parameter Request (0x12)     |
| 6   | 09:45:37 | Host→Tx   | Parameters for Modbus Slave 2          | Modbus Parameter (0x11)             |
| 7   |          | Tx→Host   | Request Parameters Slave 3             | Modbus Parameter Request (0x12)     |
| 8   | 09:45:38 | Host→Tx   | Parameters for Modbus Slave 3          | Modbus Parameter (0x11)             |
| 9   |          | Tx→Host   | Request Parameters Slave 4             | Modbus Parameter Request (0x12)     |
| 10  | 09:45:40 | Host→Tx   | Parameters for Modbus Slave 4          | Modbus Parameter (0x11)             |
| 11  |          | Tx→Host   | Request Parameters Slave 5             | Modbus Parameter Request (0x12)     |
| 12  | 09:45:41 | Host→Tx   | Parameters for Modbus Slave 5          | Modbus Parameter (0x11)             |
| 13  |          | Tx→Host   | Request Parameters Slave 6             | Modbus Parameter Request (0x12)     |
| 14  | 09:45:42 | Host→Tx   | Parameters for Modbus Slave 6          | Modbus Parameter (0x11)             |
| 15  |          | Tx→Host   | Request Parameters Slave 7             | Modbus Parameter Request (0x12)     |
| 16  | 09:45:43 | Host→Tx   | Parameters for Modbus Slave 7          | Modbus Parameter (0x11)             |
| 17  |          | Tx→Host   | Request Parameters Slave 8             | Modbus Parameter Request (0x12)     |
| 18  | 09:45:45 | Host→Tx   | Parameters for Modbus Slave 8          | Modbus Parameter (0x11)             |
| 19  |          | Tx→Host   | Request Parameters Slave 9             | Modbus Parameter Request (0x12)     |

## Enless Data Transmission Message Format

|    |          |         |                                   |                                 |
|----|----------|---------|-----------------------------------|---------------------------------|
| 20 | 09:45:46 | Host→Tx | Parameters for Modbus Slave 9     | Modbus Parameter (0x11)         |
| 21 |          | Tx→Host | Request Parameters Slave 10       | Modbus Parameter Request (0x12) |
| 22 | 09:45:47 | Host→Tx | Parameters for Modbus Slave 10    | Modbus Parameter (0x11)         |
| 23 | 09:45:49 | Tx→Host | Acknowledgement Received From Tx. | Installation ACK (0x12)         |
| 24 | 09:45:49 | Host→Tx | RSSI response Sent to Tx.         | RSSI Response (0x15)            |
| 25 | 09:45:49 | Tx→Host | RSSI 1 Message Received From Tx.  | Field Installation RSSI (0x14)  |
| 26 | 09:45:50 | Host→Tx | RSSI 1 response Sent to Tx.       | RSSI Response (0x15)            |
| 27 | 09:45:51 | Tx→Host | RSSI 2 Message Received From Tx.  | Field Installation RSSI (0x14)  |
| 28 | 09:45:51 | Host→Tx | RSSI 2 response Sent to Tx.       | RSSI Response (0x15)            |
| 29 | 09:45:52 | Tx→Host | RSSI 3 Message Received From Tx.  | Field Installation RSSI (0x14)  |
| 30 | 09:45:52 | Host→Tx | RSSI 3 response Sent to Tx.       | RSSI Response (0x15)            |
| 31 | 09:45:53 | Tx→Host | RSSI 4 Message Received From Tx.  | Field Installation RSSI (0x14)  |
| 22 | 09:45:53 | Host→Tx | RSSI 4 response Sent to Tx.       | RSSI Response (0x15)            |
| 33 | 09:45:54 | Tx→Host | Installation Successful           | Installation Successful (0x16)  |
| 44 |          | Tx→Host | Installation Successful           | Installation Successful (0x16)  |

A Complete Installation Packet Sequence for the EMT (over-the-air) can be seen below (with the packet types highlighted in Red:

```
09:45.43 681544AE0C0200102001077A010000002F2F0F7F10052E16
09:45.44 682A4424487856341201077A280000002F2F0F7F11050B2010000200050001004020000003E800003E8013B16
09:45.44 681644AE0C0200102001077A020000002F2F0F7F1205002E16
09:45.45 68264424487856341201077A280000002F2F0F7F11050B2010000201050001004020658A03E8053D16
09:45.45 681644AE0C0200102001077A030000002F2F0F7F1205012D16
09:45.46 68264424487856341201077A280000002F2F0F7F11050B2010000202050001004020668A07D0053F16
09:45.46 681644AE0C0200102001077A040000002F2F0F7F1205022E16
09:45.47 68264424487856341201077A280000002F2F0F7F11050B2010000203050001004020678A0BB8053F16
09:45.48 681644AE0C0200102001077A050000002F2F0F7F1205032C16
09:45.48 68264424487856341201077A280000002F2F0F7F11050B2010000204050001004020688A0FA0053316
09:45.49 681644AE0C0200102001077A060000002F2F0F7F1205042416
09:45.50 68264424487856341201077A280000002F2F0F7F11050B2010000205050001004020698A1388053616
09:45.50 681644AE0C0200102001077A070000002F2F0F7F1205052A16
09:45.51 68264424487856341201077A280000002F2F0F7F11050B20100002060500010040206A8A1770054116
09:45.51 681644AE0C0200102001077A080000002F2F0F7F1205062B16
09:45.52 68264424487856341201077A280000002F2F0F7F11050B20100002070500010040206B8A1B58053F16
09:45.52 681644AE0C0200102001077A090000002F2F0F7F1205072D16
09:45.53 68264424487856341201077A280000002F2F0F7F11050B20100002080500010040206C8A1F40053E16
09:45.54 681644AE0C0200102001077A0A0000002F2F0F7F1205082C16
09:45.54 68264424487856341201077A280000002F2F0F7F11050B20100002090500010040206D8A2328053E16
09:45.55 681644AE0C0200102001077A0B0000002F2F0F7F1205092C16
09:45.56 68264424487856341201077A280000002F2F0F7F11050B201000020A0500010040206E8A2710053F16
09:45.56 681644AE0C0200102001077A0C0000002F2F0F7F12050A2D16
09:45.57 681B4424487856341201077A280000002F2F0F7F15050120100002203A16
09:45.58 681644AE0C0200102001077A0D0000002F2F0F7F1405022E16
09:45.59 681B4424487856341201077A2A0000002F2F0F7F15050220100002233816
09:45.59 681644AE0C0200102001077A0E0000002F2F0F7F1405032916
09:46.00 681B4424487856341201077A2B0000002F2F0F7F15050320100002263216
09:46.00 681644AE0C0200102001077A0F0000002F2F0F7F1405042916
```

09:46.01 681B4424487856341201077A2C0000002F2F0F7F15050420100002263416  
09:46.01 681644AE0C0200102001077A100000002F2F0F7F1405052916  
09:46.02 681B4424487856341201077A2D0000002F2F0F7F15050520100002263716  
09:46.02 681544AE0C0200102001077A110000002F2F0F7F16052B16  
09:46.03 681544AE0C0200102001077A120000002F2F0F7F16052F16

### 3.1 Installation Request

| Message ID | Device Type |
|------------|-------------|
| 0x10       | 0x01 – 0x27 |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:
  1. 0x01 is the Temperature Transmitter
  2. 0x02 is the Temperature and Humidity Transmitter
  3. 0x03 is the PT100 Temperature Device
  4. 0x04 is Pulse Transmitter Device
  5. 0x05 Energy Meter Transmitter Device
  6. 0x07 Modbus Receiver
  7. 0x23 Contact Transmitter Device
  8. 0x24 CO2 Transmitter Device
  9. 0x25 4-20mA Analog Transmitter Device
  10. 0x26 0-5V Analog Transmitter Device
  11. 0x27 0-10V Analog Transmitter Device

## Enless Data Transmission Message Format

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A typical packet (over-the-air) breakdown is shown below:

| INSTALLATION REQUEST |                                                  |       |              |      |     |     |       |
|----------------------|--------------------------------------------------|-------|--------------|------|-----|-----|-------|
| Data Frame           | 681544AE0C9900901001077A010000002F2F0F7F10044D16 |       |              |      |     |     |       |
|                      | Start                                            | Count | HEX VALUE    | DEC  |     |     |       |
| Start Byte           | 1                                                | 2     | 68           |      |     |     |       |
| Length Byte          | 3                                                | 2     | 15           | 21   |     |     |       |
| C Field              | 5                                                | 2     | 44           |      |     |     |       |
| M Field              | 7                                                | 4     | AE0C         |      |     |     |       |
| A Field              | 11                                               | 12    | 990090100107 |      |     |     |       |
| CI Control           | 23                                               | 2     | 7A           |      |     |     |       |
| Access Number        | 25                                               | 2     | 01           |      |     |     |       |
| Status               | 27                                               | 2     | 00           |      |     |     |       |
| Config Word          | 29                                               | 4     | 0000         |      |     |     |       |
| AES Verify           | 33                                               | 4     | 2F2F         |      |     |     |       |
| DIF                  | 37                                               | 2     | 0F           |      |     |     |       |
| VIF                  | 39                                               | 2     | 7F           |      |     |     |       |
|                      |                                                  |       |              |      | HEX | DEC |       |
| Data                 | 41                                               | 4     | 1004         | ID   | 10  | 16  |       |
|                      |                                                  |       |              | Type | 04  | 4   |       |
|                      |                                                  |       |              |      |     |     |       |
| RSSI                 | 45                                               | 2     | 4D           |      | 4D  | 77  | -38.5 |
| Stop                 | 47                                               | 2     | 16           |      |     |     |       |

### 3.2 TX\_TEMP Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x01 - Temperature Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.



## Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| TEMPERATURE INSTALLATION PACKET |                                                                        |       |                            |               |          |       |
|---------------------------------|------------------------------------------------------------------------|-------|----------------------------|---------------|----------|-------|
| Data Frame                      | 68204424487856341201077A1E0000002F2F0F7F110101105000240500010040204616 |       |                            |               |          |       |
|                                 | Start                                                                  | Count | HEX VALUE                  | DEC           |          |       |
| Start Byte                      | 1                                                                      | 2     | 68                         |               |          |       |
| Length Byte                     | 3                                                                      | 2     | 20                         | 32            |          |       |
| C Field                         | 5                                                                      | 2     | 44                         |               |          |       |
| M Field                         | 7                                                                      | 4     | 2448                       |               |          |       |
| A Field                         | 11                                                                     | 12    | 785634120107               |               |          |       |
| CI Control                      | 23                                                                     | 2     | 7A                         |               |          |       |
| Access Number                   | 25                                                                     | 2     | 1E                         |               |          |       |
| Status                          | 27                                                                     | 2     | 00                         |               |          |       |
| Config Word                     | 29                                                                     | 4     | 0000                       |               |          |       |
| AES Verify                      | 33                                                                     | 4     | 2F2F                       |               |          |       |
| DIF                             | 37                                                                     | 2     | 0F                         |               |          |       |
| VIF                             | 39                                                                     | 2     | 7F                         |               |          |       |
|                                 |                                                                        |       |                            | HEX           | DEC      |       |
| Data                            | 41                                                                     | 26    | 11010110500024050001004020 | ID            | 11       | 17    |
|                                 |                                                                        |       |                            | Device Type   | 01       | 1     |
|                                 |                                                                        |       |                            | Frame Counter | 01       | 1     |
|                                 |                                                                        |       |                            | TX ID         | 10500024 |       |
|                                 |                                                                        |       |                            | Tx Interval   | 05       | 5     |
|                                 |                                                                        |       |                            | Re-Try Count  | 00       | 0     |
|                                 |                                                                        |       |                            | Receiver ID   | 01004020 |       |
| RSSI                            | 67                                                                     | 2     | 46                         |               | 46       | 70    |
| Stop                            | 69                                                                     | 2     | 16                         |               |          | -35.0 |

### 3.3 TX\_TEMP\_HUM Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x02 - Temperature and Humidity Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min.
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.

## Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| TEMPERATURE HUMIDITY TRANSMITTER INSTALLATION PACKET |                                                                        |       |                            |               |          |     |       |
|------------------------------------------------------|------------------------------------------------------------------------|-------|----------------------------|---------------|----------|-----|-------|
| Data Frame                                           | 68204424487856341201077A1E0000002F2F0F7F110201105000110300010040204616 |       |                            |               |          |     |       |
|                                                      | Start                                                                  | Count | HEX VALUE                  | DEC           |          |     |       |
| Start Byte                                           | 1                                                                      | 2     | 68                         |               |          |     |       |
| Length Byte                                          | 3                                                                      | 2     | 20                         | 32            |          |     |       |
| C Field                                              | 5                                                                      | 2     | 44                         |               |          |     |       |
| M Field                                              | 7                                                                      | 4     | 2448                       |               |          |     |       |
| A Field                                              | 11                                                                     | 12    | 785634120107               |               |          |     |       |
| CI Control                                           | 23                                                                     | 2     | 7A                         |               |          |     |       |
| Access Number                                        | 25                                                                     | 2     | 1E                         |               |          |     |       |
| Status                                               | 27                                                                     | 2     | 00                         |               |          |     |       |
| Config Word                                          | 29                                                                     | 4     | 0000                       |               |          |     |       |
| AES Verify                                           | 33                                                                     | 4     | 2F2F                       |               |          |     |       |
| DIF                                                  | 37                                                                     | 2     | 0F                         |               |          |     |       |
| VIF                                                  | 39                                                                     | 2     | 7F                         |               |          |     |       |
|                                                      |                                                                        |       |                            |               | HEX      | DEC |       |
| Data                                                 | 41                                                                     | 26    | 11020110500011030001004020 | ID            | 11       | 17  |       |
|                                                      |                                                                        |       |                            | Device Type   | 02       | 2   |       |
|                                                      |                                                                        |       |                            | Frame Counter | 01       | 1   |       |
|                                                      |                                                                        |       |                            | TX ID         | 10500011 |     |       |
|                                                      |                                                                        |       |                            | Tx Interval   | 03       | 3   |       |
|                                                      |                                                                        |       |                            | Re-Try Count  | 00       | 0   |       |
|                                                      |                                                                        |       |                            | Receiver ID   | 01004020 |     |       |
| RSSI                                                 | 67                                                                     | 2     | 46                         |               | 46       | 70  | -35.0 |
| Stop                                                 | 69                                                                     | 2     | 16                         |               |          |     |       |

### 3.4 TX\_TEMP\_PT100 Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x03 - PT100 Temperature
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.

## Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| PT100 TRANSMITTER INSTALLATION PACKET |                                                                        |       |                            |               |          |          |
|---------------------------------------|------------------------------------------------------------------------|-------|----------------------------|---------------|----------|----------|
| Data Frame                            | 68204424487856341201077A1E0000002F2F0F7F110301105000860500010040204616 |       |                            |               |          |          |
|                                       | Start                                                                  | Count | HEX VALUE                  | DEC           |          |          |
| Start Byte                            | 1                                                                      | 2     | 68                         |               |          |          |
| Length Byte                           | 3                                                                      | 2     | 20                         | 32            |          |          |
| C Field                               | 5                                                                      | 2     | 44                         |               |          |          |
| M Field                               | 7                                                                      | 4     | 2448                       |               |          |          |
| A Field                               | 11                                                                     | 12    | 785634120107               |               |          |          |
| CI Control                            | 23                                                                     | 2     | 7A                         |               |          |          |
| Access Number                         | 25                                                                     | 2     | 1E                         |               |          |          |
| Status                                | 27                                                                     | 2     | 00                         |               |          |          |
| Config Word                           | 29                                                                     | 4     | 0000                       |               |          |          |
| AES Verify                            | 33                                                                     | 4     | 2F2F                       |               |          |          |
| DIF                                   | 37                                                                     | 2     | 0F                         |               |          |          |
| VIF                                   | 39                                                                     | 2     | 7F                         |               |          |          |
|                                       |                                                                        |       |                            | HEX           | DEC      |          |
| Data                                  | 41                                                                     | 26    | 11030110500086050001004020 | ID            | 11       | 17       |
|                                       |                                                                        |       |                            | Device Type   | 03       | 3        |
|                                       |                                                                        |       |                            | Frame Counter | 01       | 1        |
|                                       |                                                                        |       |                            | TX ID         | 10500086 | 10900099 |
|                                       |                                                                        |       |                            | Tx Interval   | 05       | 5        |
|                                       |                                                                        |       |                            | Re-Try Count  | 00       | 0        |
|                                       |                                                                        |       |                            | Receiver ID   | 01004020 |          |
| RSSI                                  | 67                                                                     | 2     | 46                         |               | 46       | 70       |
| Stop                                  | 69                                                                     | 2     | 16                         |               |          | -35.0    |

### 3.5 TX\_PULSE Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x04 is Pulse Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below:

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 | Pulse1 Count D0 | Pulse1 Count D1 | Pulse1 Count D2 | Pulse1 Count D3 | Pulse2 Count D0 | Pulse2 Count D1 | Pulse Count D2 | Pulse1 Count D3 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|-----------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                | 0x00            | 0x00            | 0x00            | 0x00            | 0x00            | 0x00            | 0x00           | 0x0             |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.
- **Pulse Count 1 and Pulse Count2** – it will be 4 byte initial count for both pulse count 1 and pulse count 2 user can set initial count from 0x00000000 to less than 0xFFFFFFFF.

## Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| PULSE TRANSMITTER INSTALLATION PACKET |                                                                                        |       |                                            |                       |          |     |       |
|---------------------------------------|----------------------------------------------------------------------------------------|-------|--------------------------------------------|-----------------------|----------|-----|-------|
| Data Frame                            | 68284424487856341201077A0A0000002F2F0F7F110401109000990500010040200000000A000000643516 |       |                                            |                       |          |     |       |
|                                       | Start                                                                                  | Count | HEX VALUE                                  | DEC                   |          |     |       |
| Start Byte                            | 1                                                                                      | 2     | 68                                         |                       |          |     |       |
| Length Byte                           | 3                                                                                      | 2     | 28                                         | 40                    |          |     |       |
| C Field                               | 5                                                                                      | 2     | 44                                         |                       |          |     |       |
| M Field                               | 7                                                                                      | 4     | 2448                                       |                       |          |     |       |
| A Field                               | 11                                                                                     | 12    | 785634120107                               |                       |          |     |       |
| CI Control                            | 23                                                                                     | 2     | 7A                                         |                       |          |     |       |
| Access Number                         | 25                                                                                     | 2     | 0A                                         |                       |          |     |       |
| Status                                | 27                                                                                     | 2     | 00                                         |                       |          |     |       |
| Config Word                           | 29                                                                                     | 4     | 0000                                       |                       |          |     |       |
| AES Verify                            | 33                                                                                     | 4     | 2F2F                                       |                       |          |     |       |
| DIF                                   | 37                                                                                     | 2     | 0F                                         |                       |          |     |       |
| VIF                                   | 39                                                                                     | 2     | 7F                                         |                       |          |     |       |
|                                       |                                                                                        |       |                                            |                       | HEX      | DEC |       |
| Data                                  | 41                                                                                     | 42    | 110401109000990500010040200000000A00000064 | ID                    | 11       | 17  |       |
|                                       |                                                                                        |       |                                            | Device Type           | 04       | 4   |       |
|                                       |                                                                                        |       |                                            | Frame Counter         | 01       | 1   |       |
|                                       |                                                                                        |       |                                            | TX ID                 | 10900099 |     |       |
|                                       |                                                                                        |       |                                            | Tx Interval           | 05       | 5   |       |
|                                       |                                                                                        |       |                                            | Re-Try Count          | 00       | 0   |       |
|                                       |                                                                                        |       |                                            | Receiver ID           | 01004020 |     |       |
|                                       |                                                                                        |       |                                            | Initial Pulse Count 1 | 0000000A | 10  |       |
|                                       |                                                                                        |       |                                            | InitialPulse Count 2  | 00000064 | 100 |       |
| RSSI                                  | 83                                                                                     | 2     | 35                                         |                       | 35       | 53  | -26.5 |
| Stop                                  | 85                                                                                     | 2     | 16                                         |                       |          |     |       |

### 3.6 TX\_CONTACT Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where: 0x23 is Contact Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below:

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                |

- **Transmit Interval** - It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.



Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| CONTACT INSTALLATION PACKET |                                                                       |       |                            |     |               |       |       |          |       |
|-----------------------------|-----------------------------------------------------------------------|-------|----------------------------|-----|---------------|-------|-------|----------|-------|
| Data Frame                  | 682044AE0C0100002001077A000000002F2F0F7F11230111111130100000000003C16 |       |                            |     |               |       |       |          |       |
|                             | Start                                                                 | Count | HEX VALUE                  | DEC |               |       |       |          |       |
| Start Byte                  | 1                                                                     | 2     | 68                         |     |               |       |       |          |       |
| Length Byte                 | 3                                                                     | 2     | 20                         | 32  |               |       |       |          |       |
| C Field                     | 5                                                                     | 2     | 44                         |     |               |       |       |          |       |
| M Field                     | 7                                                                     | 4     | AE0C                       |     |               |       |       |          |       |
| A Field                     | 11                                                                    | 12    | 010000200107               |     |               |       |       |          |       |
| CI Control                  | 23                                                                    | 2     | 7A                         |     |               |       |       |          |       |
| Access Number               | 25                                                                    | 2     | 00                         |     |               |       |       |          |       |
| Status                      | 27                                                                    | 2     | 00                         |     |               |       |       |          |       |
| Config Word                 | 29                                                                    | 4     | 0000                       |     |               |       |       |          |       |
| AES Verify                  | 33                                                                    | 4     | 2F2F                       |     |               |       |       |          |       |
| DIF                         | 37                                                                    | 2     | 0F                         |     |               |       |       |          |       |
| VIF                         | 39                                                                    | 2     | 7F                         |     |               |       |       |          |       |
| Data                        | 41                                                                    | 26    | 11230111111130100000000000 |     | ID            | Start | Count | HEX      | DEC   |
|                             |                                                                       |       |                            |     | Device Type   | 41    | 2     | 11       | 17    |
|                             |                                                                       |       |                            |     | Frame Counter | 43    | 2     | 23       | 35    |
|                             |                                                                       |       |                            |     | TX ID         | 45    | 2     | 01       | 1     |
|                             |                                                                       |       |                            |     | Tx Interval   | 47    | 8     | 11111113 |       |
|                             |                                                                       |       |                            |     | Re-Try Count  | 55    | 2     | 01       | 1     |
|                             |                                                                       |       |                            |     | Receiver ID   | 57    | 2     | 00       | 0     |
|                             |                                                                       |       |                            |     |               | 59    | 8     | 00000000 |       |
| RSSI                        | 67                                                                    | 2     | 3C                         |     |               |       |       | 3C       | 60    |
| Stop                        | 69                                                                    | 2     | 16                         |     |               |       |       |          | -30.0 |

### 3.7 TX\_CO2 Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x24 is CO2 Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below:

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.

Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| CO2 INSTALLATION PACKET |                                                                        |       |                            |     |               |       |       |          |       |
|-------------------------|------------------------------------------------------------------------|-------|----------------------------|-----|---------------|-------|-------|----------|-------|
| Data Frame              | 682044AE0C0100002001077A000000002F2F0F7F112401104000010101000000003A16 |       |                            |     |               |       |       |          |       |
|                         | Start                                                                  | Count | HEX VALUE                  | DEC |               |       |       |          |       |
| Start Byte              | 1                                                                      | 2     | 68                         |     |               |       |       |          |       |
| Length Byte             | 3                                                                      | 2     | 20                         | 32  |               |       |       |          |       |
| C Field                 | 5                                                                      | 2     | 44                         |     |               |       |       |          |       |
| M Field                 | 7                                                                      | 4     | AE0C                       |     |               |       |       |          |       |
| A Field                 | 11                                                                     | 12    | 010000200107               |     |               |       |       |          |       |
| CI Control              | 23                                                                     | 2     | 7A                         |     |               |       |       |          |       |
| Access Number           | 25                                                                     | 2     | 00                         |     |               |       |       |          |       |
| Status                  | 27                                                                     | 2     | 00                         |     |               |       |       |          |       |
| Config Word             | 29                                                                     | 4     | 0000                       |     |               |       |       |          |       |
| AES Verify              | 33                                                                     | 4     | 2F2F                       |     |               |       |       |          |       |
| DIF                     | 37                                                                     | 2     | 0F                         |     |               |       |       |          |       |
| VIF                     | 39                                                                     | 2     | 7F                         |     |               |       |       |          |       |
| Data                    | 41                                                                     | 26    | 11240110400001010100000000 |     | ID            | Start | Count | HEX      | DEC   |
|                         |                                                                        |       |                            |     | Device Type   | 41    | 2     | 11       | 17    |
|                         |                                                                        |       |                            |     | Frame Counter | 43    | 2     | 24       | 36    |
|                         |                                                                        |       |                            |     |               | 45    | 2     | 01       | 1     |
|                         |                                                                        |       |                            |     | TX ID         | 47    | 8     | 10400001 |       |
|                         |                                                                        |       |                            |     | Tx Interval   | 55    | 2     | 01       | 1     |
|                         |                                                                        |       |                            |     | Re-Try Count  | 57    | 2     | 01       | 1     |
|                         |                                                                        |       |                            |     | Receiver ID   | 59    | 8     | 00000000 |       |
| RSSI                    | 67                                                                     | 2     | 3A                         |     |               |       |       | 3A       | 58    |
| Stop                    | 69                                                                     | 2     | 16                         |     |               |       |       |          | -29.0 |

### 3.8 TX\_4to20 Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x25 is 4-20mA Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below:

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 | Sensor Power Time D0 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|----------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                | 0x11                 |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.
- **Sensor Power Time (D0)** – one byte and denotes how many seconds the sensor will be powered for before transmission.

Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| 4-20mA INSTALLATION PACKET |                                                                         |       |                             |  |               |       |       |          |       |
|----------------------------|-------------------------------------------------------------------------|-------|-----------------------------|--|---------------|-------|-------|----------|-------|
| Data Frame                 | 682144AE0C0100002001077A000000002F2F0F7F1125011111112010100000000023A16 |       |                             |  |               |       |       |          |       |
|                            | Start                                                                   | Count | HEX VALUE                   |  | DEC           |       |       |          |       |
| Start Byte                 | 1                                                                       | 2     | 68                          |  |               |       |       |          |       |
| Length Byte                | 3                                                                       | 2     | 21                          |  | 33            |       |       |          |       |
| C Field                    | 5                                                                       | 2     | 44                          |  |               |       |       |          |       |
| M Field                    | 7                                                                       | 4     | AE0C                        |  |               |       |       |          |       |
| A Field                    | 11                                                                      | 12    | 010000200107                |  |               |       |       |          |       |
| CI Control                 | 23                                                                      | 2     | 7A                          |  |               |       |       |          |       |
| Access Number              | 25                                                                      | 2     | 00                          |  |               |       |       |          |       |
| Status                     | 27                                                                      | 2     | 00                          |  |               |       |       |          |       |
| Config Word                | 29                                                                      | 4     | 0000                        |  |               |       |       |          |       |
| AES Verify                 | 33                                                                      | 4     | 2F2F                        |  |               |       |       |          |       |
| DIF                        | 37                                                                      | 2     | 0F                          |  |               |       |       |          |       |
| VIF                        | 39                                                                      | 2     | 7F                          |  |               |       |       |          |       |
|                            |                                                                         |       |                             |  |               | Start | Count | HEX      | DEC   |
| Data                       | 41                                                                      | 28    | 112501111111201010000000002 |  | ID            | 41    | 2     | 11       | 17    |
|                            |                                                                         |       |                             |  | Device Type   | 43    | 2     | 25       | 37    |
|                            |                                                                         |       |                             |  | Frame Counter | 45    | 2     | 01       | 1     |
|                            |                                                                         |       |                             |  | TX ID         | 47    | 8     | 11111112 |       |
|                            |                                                                         |       |                             |  | Tx Interval   | 55    | 2     | 01       | 1     |
|                            |                                                                         |       |                             |  | Re-Try Count  | 57    | 2     | 01       | 1     |
|                            |                                                                         |       |                             |  | Receiver ID   | 59    | 8     | 00000000 |       |
|                            |                                                                         |       |                             |  | Wake Period   | 67    | 2     | 02       | 2     |
| RSSI                       | 69                                                                      | 2     | 3A                          |  |               |       |       | 3A       | 58    |
| Stop                       | 71                                                                      | 2     | 16                          |  |               |       |       |          | -29.0 |

### 3.9 TX\_0to5 Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x26 is 0-5V Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below:

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 | Sensor Power Time D0 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|----------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                | 0x11                 |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.
- **Sensor Power Time (D0)** – one byte and denotes how many seconds the sensor will be powered for before transmission.

Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| 0-5V INSTALLATION PACKET |                                                                          |       |                              |     |               |    |   |          |       |
|--------------------------|--------------------------------------------------------------------------|-------|------------------------------|-----|---------------|----|---|----------|-------|
| Data Frame               | 682144AE0C0100002001077A000000002F2F0F7F11260111111110010100000000043A16 |       |                              |     |               |    |   |          |       |
|                          | Start                                                                    | Count | HEX VALUE                    | DEC |               |    |   |          |       |
| Start Byte               | 1                                                                        | 2     | 68                           |     |               |    |   |          |       |
| Length Byte              | 3                                                                        | 2     | 21                           | 33  |               |    |   |          |       |
| C Field                  | 5                                                                        | 2     | 44                           |     |               |    |   |          |       |
| M Field                  | 7                                                                        | 4     | AEOC                         |     |               |    |   |          |       |
| A Field                  | 11                                                                       | 12    | 010000200107                 |     |               |    |   |          |       |
| CI Control               | 23                                                                       | 2     | 7A                           |     |               |    |   |          |       |
| Access Number            | 25                                                                       | 2     | 00                           |     |               |    |   |          |       |
| Status                   | 27                                                                       | 2     | 00                           |     |               |    |   |          |       |
| Config Word              | 29                                                                       | 4     | 0000                         |     |               |    |   |          |       |
| AES Verify               | 33                                                                       | 4     | 2F2F                         |     |               |    |   |          |       |
| DIF                      | 37                                                                       | 2     | 0F                           |     |               |    |   |          |       |
| VIF                      | 39                                                                       | 2     | 7F                           |     |               |    |   |          |       |
| Data                     | 41                                                                       | 28    | 1126011111111001010000000004 |     | ID            | 41 | 2 | 11       | 17    |
|                          |                                                                          |       |                              |     | Device Type   | 43 | 2 | 26       | 38    |
|                          |                                                                          |       |                              |     | Frame Counter | 45 | 2 | 01       | 1     |
|                          |                                                                          |       |                              |     | TX ID         | 47 | 8 | 11111110 |       |
|                          |                                                                          |       |                              |     | Tx Interval   | 55 | 2 | 01       | 1     |
|                          |                                                                          |       |                              |     | Re-Try Count  | 57 | 2 | 01       | 1     |
|                          |                                                                          |       |                              |     | Receiver ID   | 59 | 8 | 00000000 |       |
|                          |                                                                          |       |                              |     | Wake Period   | 67 | 2 | 04       | 4     |
| RSSI                     | 69                                                                       | 2     | 3A                           |     |               |    |   | 3A       | 58    |
| Stop                     | 71                                                                       | 2     | 16                           |     |               |    |   |          | -29.0 |

### 3.10 TX\_0to10 Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x27 is 0-10V Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below:

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 | Sensor Power Time D0 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|----------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                | 0x11                 |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.
- **Sensor Power Time (D0)** – one byte and denotes how many seconds the sensor will be powered for before transmission.



Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| 0-10V INSTALLATION PACKET |                                                                         |       |                              |  |               |       |       |          |     |       |
|---------------------------|-------------------------------------------------------------------------|-------|------------------------------|--|---------------|-------|-------|----------|-----|-------|
| Data Frame                | 682144AE0C0100002001077A000000002F2F0F7F1127011111111101010000000053916 |       |                              |  |               |       |       |          |     |       |
|                           | Start                                                                   | Count | HEX VALUE                    |  | DEC           |       |       |          |     |       |
| Start Byte                | 1                                                                       | 2     | 68                           |  |               |       |       |          |     |       |
| Length Byte               | 3                                                                       | 2     | 21                           |  | 33            |       |       |          |     |       |
| C Field                   | 5                                                                       | 2     | 44                           |  |               |       |       |          |     |       |
| M Field                   | 7                                                                       | 4     | AE0C                         |  |               |       |       |          |     |       |
| A Field                   | 11                                                                      | 12    | 010000200107                 |  |               |       |       |          |     |       |
| CI Control                | 23                                                                      | 2     | 7A                           |  |               |       |       |          |     |       |
| Access Number             | 25                                                                      | 2     | 00                           |  |               |       |       |          |     |       |
| Status                    | 27                                                                      | 2     | 00                           |  |               |       |       |          |     |       |
| Config Word               | 29                                                                      | 4     | 0000                         |  |               |       |       |          |     |       |
| AES Verify                | 33                                                                      | 4     | 2F2F                         |  |               |       |       |          |     |       |
| DIF                       | 37                                                                      | 2     | 0F                           |  |               |       |       |          |     |       |
| VIF                       | 39                                                                      | 2     | 7F                           |  |               |       |       |          |     |       |
|                           |                                                                         |       |                              |  |               | Start | Count | HEX      | DEC |       |
| Data                      | 41                                                                      | 28    | 1127011111111101010000000005 |  | ID            | 41    | 2     | 11       | 17  |       |
|                           |                                                                         |       |                              |  | Device Type   | 43    | 2     | 27       | 39  |       |
|                           |                                                                         |       |                              |  | Frame Counter | 45    | 2     | 01       | 1   |       |
|                           |                                                                         |       |                              |  | TX ID         | 47    | 8     | 11111111 |     |       |
|                           |                                                                         |       |                              |  | Tx Interval   | 55    | 2     | 01       | 1   |       |
|                           |                                                                         |       |                              |  | Re-Try Count  | 57    | 2     | 01       | 1   |       |
|                           |                                                                         |       |                              |  | Receiver ID   | 59    | 8     | 00000000 |     |       |
|                           |                                                                         |       |                              |  | Wake Period   | 67    | 2     | 05       | 5   |       |
| RSSI                      | 69                                                                      | 2     | 39                           |  |               |       |       | 39       | 57  | -28.5 |
| Stop                      | 71                                                                      | 2     | 16                           |  |               |       |       |          |     |       |

### 3.11 TX\_ENERGY Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x05 Energy Meter Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below table

#### 3.11.1 TX\_ENERGY Pulse Installation

| Message Number | Transmit Interval | Retry Count | Recvr Address D0 | Recvr Address D1 | Recvr Address D2 | Recvr Address D3 | Pulse1 Count D0 | Pulse1 Count D1 | Pulse1 Count D2 | Pulse1 Count D3 | Pulse2 Count D0 | Pulse2 Count D1 | Pulse Count D2 | Pulse1 Count D3 |
|----------------|-------------------|-------------|------------------|------------------|------------------|------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|-----------------|
| 0x00           | 0x03 – 0xFF       | 0x00 – 0x01 | 0x44             | 0x33             | 0x22             | 0x11             | 0x00            | 0x00            | 0x00            | 0x00            | 0x00            | 0x00            | 0x00           | 0x00            |

- **Message Number** – it will be always 0x00 for Energy meter pulse counter device.
- **Transmit Interval** – It will be 0x03 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 3 to 255 minute. If set to below 0x03, the transmitter will default to 3min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of transmitter address. In given example 0x11223344.
- **Pulse Count 1 and Pulse Count2** – it will be 4 byte initial count for both pulse count 1 and pulse count 2 user can set initial count from 0x00000000 to less than 0xFFFFFFFF.

## Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| EMT PULSE TRANSMITTER INSTALLATION PACKET |                                                                                            |       |                                                |                       |          |       |
|-------------------------------------------|--------------------------------------------------------------------------------------------|-------|------------------------------------------------|-----------------------|----------|-------|
| Data Frame                                | 682A4424487856341201077A280000002F2F0F7F11050B2010000200050001004020000003E8000003E8013B16 |       |                                                |                       |          |       |
|                                           | Start                                                                                      | Count | HEX VALUE                                      | DEC                   |          |       |
| Start Byte                                | 1                                                                                          | 2     | 68                                             |                       |          |       |
| Length Byte                               | 3                                                                                          | 2     | 2A                                             | 42                    |          |       |
| C Field                                   | 5                                                                                          | 2     | 44                                             |                       |          |       |
| M Field                                   | 7                                                                                          | 4     | 2448                                           |                       |          |       |
| A Field                                   | 11                                                                                         | 12    | 785634120107                                   |                       |          |       |
| CI Control                                | 23                                                                                         | 2     | 7A                                             |                       |          |       |
| Access Number                             | 25                                                                                         | 2     | 28                                             |                       |          |       |
| Status                                    | 27                                                                                         | 2     | 00                                             |                       |          |       |
| Config Word                               | 29                                                                                         | 4     | 0000                                           |                       |          |       |
| AES Verify                                | 33                                                                                         | 4     | 2F2F                                           |                       |          |       |
| DIF                                       | 37                                                                                         | 2     | 0F                                             |                       |          |       |
| VIF                                       | 39                                                                                         | 2     | 7F                                             |                       |          |       |
|                                           |                                                                                            |       |                                                |                       | HEX      | DEC   |
| Data                                      | 41                                                                                         | 46    | 11050B2010000200050001004020000003E8000003E801 | ID                    | 11       | 17    |
|                                           |                                                                                            |       |                                                | Device Type           | 05       | 5     |
|                                           |                                                                                            |       |                                                | Frame Counter         | 0B       | 11    |
|                                           |                                                                                            |       |                                                | TX ID                 | 20100002 |       |
|                                           |                                                                                            |       |                                                | Message Num           | 00       | 0     |
|                                           |                                                                                            |       |                                                | Tx Interval           | 05       | 5     |
|                                           |                                                                                            |       |                                                | Re-Try Count          | 00       | 0     |
|                                           |                                                                                            |       |                                                | Receiver ID           | 01004020 |       |
|                                           |                                                                                            |       |                                                | Initial Pulse Count 1 | 000003E8 | 1000  |
|                                           |                                                                                            |       |                                                | Initial Pulse Count 2 | 000003E8 | 1000  |
|                                           |                                                                                            |       |                                                |                       |          |       |
| RSSI                                      | 87                                                                                         | 2     | 3B                                             |                       | 3B       | 59    |
| Stop                                      | 89                                                                                         | 2     | 16                                             |                       |          | -29.5 |

### 3.11.2 *TX\_ENERGY Modbus Slave Installation*

| Message Number | Transmit Interval | Retry Count | Recvr Address D0 | Recvr Address D1 | Recvr Address D2 | Recvr Address D3 | Modbus Slave ID | UART Setting | First Register Address D0 | First Register Address D1 | No. Of Register to Read | Modbus Readout Interval |
|----------------|-------------------|-------------|------------------|------------------|------------------|------------------|-----------------|--------------|---------------------------|---------------------------|-------------------------|-------------------------|
| 0x01 – 0x0A    | 0x05 – 0xFF       | 0x00 – 0x01 | 0x44             | 0x33             | 0x22             | 0x11             | 0x64            | 0x83         | 0x52                      | 0xC5                      | 0x02                    | 0x05                    |

- **Message Number** – it will be 0x01 to 0x0A for Energy meter Modbus slave device parameter message number. Number of message depending on slave device entered by user in field installation tool cvs file.
- **Transmit Interval** – It will be 0x05 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 5 to 255 minute.
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of transmitter address. In given example 0x11223344.
- **Modbus Slave ID** – It will be unsigned one byte data of Modbus slave Address. In the given example is 0x64 which is 100 (decimal) address of slave device is.
- **UART setting** – It will be unsigned one byte encoded data of slave device communication baud rate. In the given example is 0x83 which is 19200-baud rate, None-parity, 2-Stopbit, 7 bit–Data bit, Parity Error Char-Ignore.

UART setting data decode example:

0x83 -> binary (MSB) 1 0 0 0 0 0 1 1 (LSB )  
D7 D6 D5 D4 D3 D2 D1 D0

LSB 3 bits (D2, D1, and D0) are consists of baud rate selection see below for more details.

D2 D1 D0  
 0 0 0 -> 2400 baud rate  
 0 0 1 -> 4800 baud rate  
 0 1 0 -> 9600 baud rate  
 0 1 1 -> 19200 baud rate  
 1 0 0 -> 38400 baud rate  
 1 0 1 -> Not Applicable  
 1 1 0 -> Not Applicable  
 1 1 1 -> Not Applicable

## Enless Data Transmission Message Format

MSB 2 bits (D7, D6) are consists of stop bit selection see below for more details.

D7 D6

0 0 -> Stop bit 1

0 1 -> Not Applicable

1 0 -> Stop bit 2

1 1 -> Not Applicable

MSB 2 bits (D5, D4) are consists of Parity bit selection see below for more details.

D5 D4

0 0 -> Parity None

0 1 -> Parity Even

1 0 -> Parity Odd

1 1 -> Not Applicable

D3 Data bit

0 – 7 Data bits

1 – 8 Data bits

A typical packet (over-the-air) breakdown is shown below:

| EMT MODBUS TRANSMITTER INSTALLATION PACKET |                                                                                    |       |                                        |                    |          |                                        |
|--------------------------------------------|------------------------------------------------------------------------------------|-------|----------------------------------------|--------------------|----------|----------------------------------------|
| Data Frame                                 | 68264424487856341201077A280000002F2F0F7F11050B2010000203050001004020678A0BB8053F16 |       |                                        |                    |          |                                        |
|                                            | Start                                                                              | Count | HEX VALUE                              | DEC                |          |                                        |
| Start Byte                                 | 1                                                                                  | 2     | 68                                     |                    |          |                                        |
| Length Byte                                | 3                                                                                  | 2     | 26                                     | 38                 |          |                                        |
| C Field                                    | 5                                                                                  | 2     | 44                                     |                    |          |                                        |
| M Field                                    | 7                                                                                  | 4     | 2448                                   |                    |          |                                        |
| A Field                                    | 11                                                                                 | 12    | 785634120107                           |                    |          |                                        |
| CI Control                                 | 23                                                                                 | 2     | 7A                                     |                    |          |                                        |
| Access Number                              | 25                                                                                 | 2     | 28                                     |                    |          |                                        |
| Status                                     | 27                                                                                 | 2     | 00                                     |                    |          |                                        |
| Config Word                                | 29                                                                                 | 4     | 0000                                   |                    |          |                                        |
| AES Verify                                 | 33                                                                                 | 4     | 2F2F                                   |                    |          |                                        |
| DIF                                        | 37                                                                                 | 2     | 0F                                     |                    |          |                                        |
| VIF                                        | 39                                                                                 | 2     | 7F                                     |                    |          |                                        |
|                                            |                                                                                    |       |                                        | HEX                | DEC      |                                        |
| Data                                       | 41                                                                                 | 38    | 11050B2010000203050001004020678A0BB805 | ID                 | 11       | 17                                     |
|                                            |                                                                                    |       |                                        | Device Type        | 05       | 5                                      |
|                                            |                                                                                    |       |                                        | Frame Counter      | 0B       | 11                                     |
|                                            |                                                                                    |       |                                        | TX ID              | 20100002 | EMT Address                            |
|                                            |                                                                                    |       |                                        | Message Num        | 03       | 3                                      |
|                                            |                                                                                    |       |                                        | Tx Interval        | 05       | 5                                      |
|                                            |                                                                                    |       |                                        | Re-Try Count       | 00       | 0                                      |
|                                            |                                                                                    |       |                                        | Receiver ID        | 01004020 |                                        |
|                                            |                                                                                    |       |                                        | Mod Slave ID       | 67       | 103                                    |
|                                            |                                                                                    |       |                                        | UART Settings      | 8A       | 138                                    |
|                                            |                                                                                    |       |                                        | First Register Add | 0BB8     | 3000                                   |
|                                            |                                                                                    |       |                                        | Num Regs.          | 05       | 5                                      |
|                                            |                                                                                    |       |                                        |                    |          | Number of Registers to Read from Slave |
| RSSI                                       | 79                                                                                 | 2     | 3F                                     |                    | 3F       | 63                                     |
| Stop                                       | 81                                                                                 | 2     | 16                                     |                    |          | -31.5                                  |

### 3.12 Installation ACK

| Message ID | Device Type | Frame Count |
|------------|-------------|-------------|
| 0x12       | 0x01 – 0xFF | 0x00-0xFF   |

- **Message ID** – It will be always 0x12 – Installation ACK
- **Device Type** – It will be 0x01 – 0xFF depending on transmitter type
- **Frame Count** – Increments to request for more data of next RSSI test depending on transmitter type

A typical packet (over-the-air) breakdown is shown below

| INSTALLATION ACKNOWLEDGEMENT PACKET |                                                    |              |                  |                    |            |            |       |
|-------------------------------------|----------------------------------------------------|--------------|------------------|--------------------|------------|------------|-------|
| <b>Data Frame</b>                   | 681644AE0C9900901001077A020000002F2F0F7F1204004C16 |              |                  |                    |            |            |       |
|                                     | <b>Start</b>                                       | <b>Count</b> | <b>HEX VALUE</b> | <b>DEC</b>         |            |            |       |
| <b>Start Byte</b>                   | 1                                                  | 2            | 68               |                    |            |            |       |
| <b>Length Byte</b>                  | 3                                                  | 2            | 16               | 22                 |            |            |       |
| <b>C Field</b>                      | 5                                                  | 2            | 44               |                    |            |            |       |
| <b>M Field</b>                      | 7                                                  | 4            | AE0C             |                    |            |            |       |
| <b>A Field</b>                      | 11                                                 | 12           | 990090100107     |                    |            |            |       |
| <b>CI Control</b>                   | 23                                                 | 2            | 7A               |                    |            |            |       |
| <b>Access Number</b>                | 25                                                 | 2            | 02               |                    |            |            |       |
| <b>Status</b>                       | 27                                                 | 2            | 00               |                    |            |            |       |
| <b>Config Word</b>                  | 29                                                 | 4            | 0000             |                    |            |            |       |
| <b>AES Verify</b>                   | 33                                                 | 4            | 2F2F             |                    |            |            |       |
| <b>DIF</b>                          | 37                                                 | 2            | 0F               |                    |            |            |       |
| <b>VIF</b>                          | 39                                                 | 2            | 7F               |                    |            |            |       |
|                                     |                                                    |              |                  |                    | <b>HEX</b> | <b>DEC</b> |       |
| <b>Data</b>                         | 41                                                 | 6            | 120400           | <b>ID</b>          | 12         | 18         |       |
|                                     |                                                    |              |                  | <b>Device Type</b> | 04         | 4          |       |
|                                     |                                                    |              |                  | <b>Frame Count</b> | 00         | 0          |       |
|                                     |                                                    |              |                  |                    |            |            |       |
| <b>RSSI</b>                         | 47                                                 | 2            | 4C               |                    | 4C         | 76         | -38.0 |
| <b>Stop</b>                         | 49                                                 | 2            | 16               |                    |            |            |       |

## 3.13 Installation RSSI Request

| Message ID | Device Type | Frame Count |
|------------|-------------|-------------|
| 0x14       | 0x01 – 0xFF | 0x00-0xFF   |

- **Message ID** – It will be always 0x14 – RSSI Request
- **Device Type** – It will be 0x01 – 0xFF depending on transmitter type
- **Frame Count** – Increments to request for more data of next RSSI test depending on transmitter type

A typical packet (over-the-air) breakdown is shown below:

| INSTALLATION RSSI REQUEST PACKET |                                                    |              |                  |                    |            |            |       |
|----------------------------------|----------------------------------------------------|--------------|------------------|--------------------|------------|------------|-------|
| <b>Data Frame</b>                | 681644AE0C9900901001077A030000002F2F0F7F1404025416 |              |                  |                    |            |            |       |
|                                  | <b>Start</b>                                       | <b>Count</b> | <b>HEX VALUE</b> | <b>DEC</b>         |            |            |       |
| <b>Start Byte</b>                | 1                                                  | 2            | 68               |                    |            |            |       |
| <b>Length Byte</b>               | 3                                                  | 2            | 16               | 22                 |            |            |       |
| <b>C Field</b>                   | 5                                                  | 2            | 44               |                    |            |            |       |
| <b>M Field</b>                   | 7                                                  | 4            | AE0C             |                    |            |            |       |
| <b>A Field</b>                   | 11                                                 | 12           | 990090100107     |                    |            |            |       |
| <b>CI Control</b>                | 23                                                 | 2            | 7A               |                    |            |            |       |
| <b>Access Number</b>             | 25                                                 | 2            | 03               |                    |            |            |       |
| <b>Status</b>                    | 27                                                 | 2            | 00               |                    |            |            |       |
| <b>Config Word</b>               | 29                                                 | 4            | 0000             |                    |            |            |       |
| <b>AES Verify</b>                | 33                                                 | 4            | 2F2F             |                    |            |            |       |
| <b>DIF</b>                       | 37                                                 | 2            | 0F               |                    |            |            |       |
| <b>VIF</b>                       | 39                                                 | 2            | 7F               |                    |            |            |       |
|                                  |                                                    |              |                  |                    | <b>HEX</b> | <b>DEC</b> |       |
| <b>Data</b>                      | 41                                                 | 6            | 140402           | <b>ID</b>          | 14         | 20         |       |
|                                  |                                                    |              |                  | <b>Device Type</b> | 04         | 4          |       |
|                                  |                                                    |              |                  | <b>Frame Count</b> | 02         | 2          |       |
|                                  |                                                    |              |                  |                    |            |            |       |
| <b>RSSI</b>                      | 47                                                 | 2            | 54               |                    | 54         | 84         | -42.0 |
| <b>Stop</b>                      | 49                                                 | 2            | 16               |                    |            |            |       |

### 3.14 RSSI Response

| Message ID | Device Type | Frame Count | Transmitter ID | RSSI RX |
|------------|-------------|-------------|----------------|---------|
| 0x15       | 0x01 – 0xFF | 0x00-0xFF   | XXXXXXXX       | 0x3A    |

- **Message ID** – It will be always 0x15 – RSSI Response
- **Device Type** – It will be 0x01 – 0xFF depending on transmitter type
- **Frame Count** – Increments to request for more data of next RSSI test depending on transmitter type
- **Transmitter ID** – ID of the transmitter that is being Installed
- **RSSI RX** – This is the received RSSI level from the transmitter. It is sent back to the transmitter as validation of the signal level between Transmitter and Receiver.

A typical packet (over-the-air) breakdown is shown below:

| RSSI RESPONSE PACKET |                                                              |              |                  |                      |            |            |
|----------------------|--------------------------------------------------------------|--------------|------------------|----------------------|------------|------------|
| <b>Data Frame</b>    | 681B4424487856341201077 000000002F2F0F7F150401109000993A3916 |              |                  |                      |            |            |
|                      | <b>Start</b>                                                 | <b>Count</b> | <b>HEX VALUE</b> | <b>DEC</b>           |            |            |
| Start Byte           | 1                                                            | 2            | 68               |                      |            |            |
| Length Byte          | 3                                                            | 2            | 1B               | 27                   |            |            |
| C Field              | 5                                                            | 2            | 44               |                      |            |            |
| M Field              | 7                                                            | 4            | 2448             |                      |            |            |
| A Field              | 11                                                           | 12           | 785634120107     |                      |            |            |
| CI Control           | 23                                                           | 2            | 7                |                      |            |            |
| Access Number        | 25                                                           | 2            | 00               |                      |            |            |
| Status               | 27                                                           | 2            | 00               |                      |            |            |
| Config Word          | 29                                                           | 4            | 0000             |                      |            |            |
| AES Verify           | 33                                                           | 4            | 2F2F             |                      |            |            |
| DIF                  | 37                                                           | 2            | 0F               |                      |            |            |
| VIF                  | 39                                                           | 2            | 7F               |                      |            |            |
|                      |                                                              |              |                  |                      | <b>HEX</b> | <b>DEC</b> |
| <b>Data</b>          | 41                                                           | 16           | 150401109000993A | <b>ID</b>            | 15         | 21         |
|                      |                                                              |              |                  | <b>Device Type</b>   | 04         | 4          |
|                      |                                                              |              |                  | <b>Frame Counter</b> | 01         | 1          |
|                      |                                                              |              |                  | <b>TX ID</b>         | 10900099   | 10900099   |
|                      |                                                              |              |                  | <b>RSSI Rx'd</b>     | 3A         | 58         |
| <b>RSSI</b>          | 57                                                           | 2            | 39               |                      | 39         | 57         |
| <b>Stop</b>          | 59                                                           | 2            | 16               |                      |            | -28.5      |



### 3.15 Installation Successful

| Message ID | Device Type | Frame Count |
|------------|-------------|-------------|
| 0x14       | 0x01 – 0xFF | 0x00-0xFF   |

- **Message ID** – It will be always 0x16 – Installation Successful
- **Device Type** – It will be 0x01 – 0xFF depending on transmitter type
- **Frame Count** – Increments to request for more data of next RSSI test depending on transmitter type

A typical packet (over-the-air) breakdown is shown below:

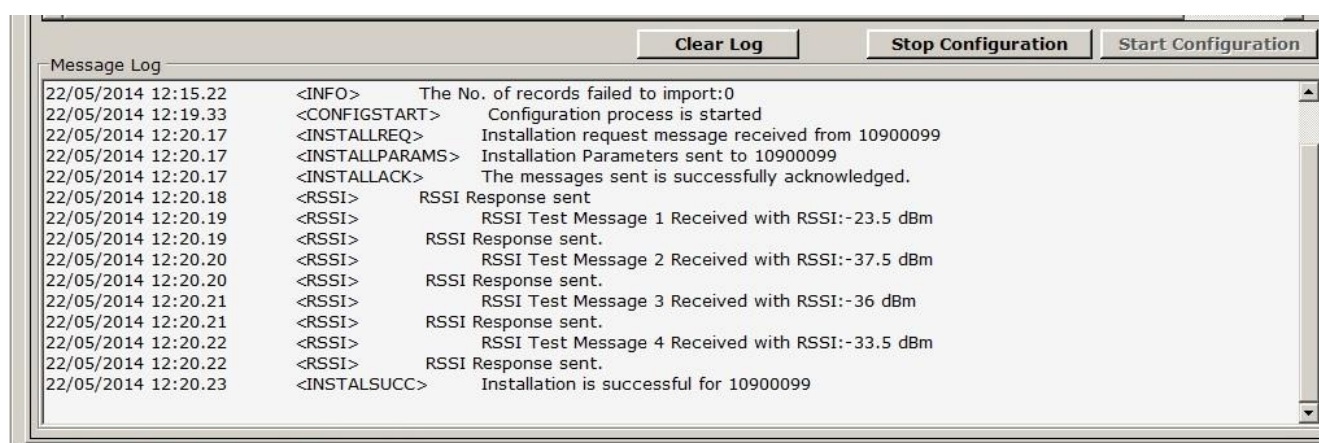
| INSTALLATION SUCCESSFUL MESSAGE |                                                  |              |                  |                    |            |            |       |
|---------------------------------|--------------------------------------------------|--------------|------------------|--------------------|------------|------------|-------|
| <b>Data Frame</b>               | 681544AE0C9900901001077A070000002F2F0F7F16044016 |              |                  |                    |            |            |       |
|                                 | <b>Start</b>                                     | <b>Count</b> | <b>HEX VALUE</b> | <b>DEC</b>         |            |            |       |
| <b>Start Byte</b>               | 1                                                | 2            | 68               |                    |            |            |       |
| <b>Length Byte</b>              | 3                                                | 2            | 15               | 21                 |            |            |       |
| <b>C Field</b>                  | 5                                                | 2            | 44               |                    |            |            |       |
| <b>M Field</b>                  | 7                                                | 4            | AE0C             |                    |            |            |       |
| <b>A Field</b>                  | 11                                               | 12           | 990090100107     |                    |            |            |       |
| <b>CI Control</b>               | 23                                               | 2            | 7A               |                    |            |            |       |
| <b>Access Number</b>            | 25                                               | 2            | 07               |                    |            |            |       |
| <b>Status</b>                   | 27                                               | 2            | 00               |                    |            |            |       |
| <b>Config Word</b>              | 29                                               | 4            | 0000             |                    |            |            |       |
| <b>AES Verify</b>               | 33                                               | 4            | 2F2F             |                    |            |            |       |
| <b>DIF</b>                      | 37                                               | 2            | 0F               |                    |            |            |       |
| <b>VIF</b>                      | 39                                               | 2            | 7F               |                    |            |            |       |
|                                 |                                                  |              |                  |                    | <b>HEX</b> | <b>DEC</b> |       |
| <b>Data</b>                     | 41                                               | 4            | 1604             | <b>ID</b>          | 16         | 22         |       |
|                                 |                                                  |              |                  | <b>Device Type</b> | 04         | 4          |       |
|                                 |                                                  |              |                  |                    |            |            |       |
| <b>RSSI</b>                     | 45                                               | 2            | 40               |                    | 40         | 64         | -32.0 |
| <b>Stop</b>                     | 47                                               | 2            | 16               |                    |            |            |       |

### 4. Field Installation Messages (with Enless Receiver)

For Field installation, the process is initiated by the transmitters. The host replies back to the messages which includes the configuration data and acknowledgements to messages. The section details the messages that are transferred between the host and the Enless Receiver.

#### Communications Settings: 19200,N,8,1

Below is a typical sequence seen on the field installation tool for all transmitters, except the Energy Meter Transmitter:



The sequence of communications is as follows:

| No. | Time     | Direction | Description                            | Message Type                  |
|-----|----------|-----------|----------------------------------------|-------------------------------|
| 1   | 12:20:17 | Tx→Host   | Installation Request Received from Tx. | Installation Request (0x10)   |
| 2   | 12:20:17 | Host→Tx   | Installation Parameters Sent to Tx.    | Installation Parameter (0x11) |
| 3   | 12:20:17 | Tx→Host   | Acknowledgement Received From Tx.      | Installation ACK (0x12)       |
| 4   | 12:20:18 | Host→Tx   | RSSI response Sent to Tx.              | RSSI Response (0x15)          |
| 5   | 12:20:19 | Tx→Host   | RSSI 1 Message Received From Tx.       | Installation RSSI (0x14)      |
| 6   | 12:20:19 | Host→Tx   | RSSI 1 response Sent to Tx.            | RSSI Response (0x15)          |
| 7   | 12:20:20 | Tx→Host   | RSSI 2 Message Received From Tx.       | Installation RSSI (0x14)      |
| 8   | 12:20:20 | Host→Tx   | RSSI 2 response Sent to Tx.            | RSSI Response (0x15)          |
| 9   | 12:20:21 | Tx→Host   | RSSI 3 Message Received From Tx.       | Installation RSSI (0x14)      |

## Enless Data Transmission Message Format

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|    |          |         |                                  |                                |
|----|----------|---------|----------------------------------|--------------------------------|
| 10 | 12:20:21 | Host→Tx | RSSI 3 response Sent to Tx.      | RSSI Response (0x15)           |
| 11 | 12:20:22 | Tx→Host | RSSI 4 Message Received From Tx. | Installation RSSI (0x14)       |
| 12 | 12:20:22 | Host→Tx | RSSI 4 response Sent to Tx.      | RSSI Response (0x15)           |
| 13 | 12:20:23 | Tx→Host | Installation Successful          | Installation Successful (0x16) |
| 14 |          | Tx→Host | Installation Successful          | Installation Successful (0x16) |

Some Complete Installation Sequence Packets (with the Enless Receiver) can be seen below (with the packet types highlighted in Red:

RX: Messages Received by Host from Enless Receiver

TX: Messages sent Out from Host to Enless Receiver

Pulse transmitter (10900099):

RX: \$68 \$15 \$44 \$AE \$0C \$99 \$00 \$90 \$10 \$01 \$07 \$7A \$01 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$10** \$04 \$41 \$16  
TX: \$1E \$7A \$49 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$04 \$01 \$10 \$90 \$00 \$99 \$05 \$00 \$01 \$00 \$40 \$20 \$00 \$00 \$0A \$00 \$00 \$00 \$64  
RX: \$68 \$16 \$44 \$AE \$0C \$99 \$00 \$90 \$10 \$01 \$07 \$7A \$02 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$12** \$04 \$00 \$44 \$16  
TX: \$11 \$7A \$49 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$15** \$04 \$01 \$10 \$90 \$00 \$99 \$43  
RX: \$68 \$16 \$44 \$AE \$0C \$99 \$00 \$90 \$10 \$01 \$07 \$7A \$03 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$14** \$04 \$02 \$44 \$16  
TX: \$11 \$7A \$4B \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$15** \$04 \$02 \$10 \$90 \$00 \$99 \$42  
RX: \$68 \$16 \$44 \$AE \$0C \$99 \$00 \$90 \$10 \$01 \$07 \$7A \$04 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$14** \$04 \$03 \$43 \$16  
TX: \$11 \$7A \$4C \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$15** \$04 \$03 \$10 \$90 \$00 \$99 \$44  
RX: \$68 \$16 \$44 \$AE \$0C \$99 \$00 \$90 \$10 \$01 \$07 \$7A \$05 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$14** \$04 \$04 \$43 \$16  
TX: \$11 \$7A \$4D \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$15** \$04 \$04 \$10 \$90 \$00 \$99 \$44  
RX: \$68 \$16 \$44 \$AE \$0C \$99 \$00 \$90 \$10 \$01 \$07 \$7A \$06 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$14** \$04 \$05 \$44 \$16  
TX: \$11 \$7A \$4E \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$15** \$04 \$05 \$10 \$90 \$00 \$99 \$42  
RX: \$68 \$15 \$44 \$AE \$0C \$99 \$00 \$90 \$10 \$01 \$07 \$7A \$07 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$16** \$04 \$43 \$16  
RX: \$68 \$15 \$44 \$AE \$0C \$99 \$00 \$90 \$10 \$01 \$07 \$7A \$08 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$16** \$04 \$40 \$16

PT100 Transmitter (10500018):

RX: \$68 \$15 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$01 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$10** \$03 \$22 \$16  
TX: \$16 \$7A \$5D \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$03 \$01 \$10 \$50 \$00 \$18 \$05 \$00 \$01 \$00 \$40 \$20  
RX: \$68 \$16 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$02 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$12** \$03 \$00 \$20 \$16  
TX: \$11 \$7A \$5D \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$15** \$03 \$01 \$10 \$50 \$00 \$18 \$2F  
RX: \$68 \$16 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$03 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$14** \$03 \$02 \$20 \$16  
TX: \$11 \$7A \$5F \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$15** \$03 \$02 \$10 \$50 \$00 \$18 \$30  
RX: \$68 \$16 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$04 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$14** \$03 \$03 \$20 \$16  
TX: \$11 \$7A \$60 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$15** \$03 \$03 \$10 \$50 \$00 \$18 \$2F  
RX: \$68 \$16 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$05 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$14** \$03 \$04 \$20 \$16  
TX: \$11 \$7A \$61 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$15** \$03 \$04 \$10 \$50 \$00 \$18 \$34  
RX: \$68 \$16 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$06 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$14** \$03 \$05 \$1F \$16  
TX: \$11 \$7A \$62 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$15** \$03 \$05 \$10 \$50 \$00 \$18 \$26  
RX: \$68 \$15 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$07 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$16** \$03 \$1F \$16  
RX: \$68 \$15 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$08 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$16** \$03 \$20 \$16

## Enless Data Transmission Message Format

For the Energy Meter Transmitter, below is the sequence:

| Message Log         |                 |                                                     |
|---------------------|-----------------|-----------------------------------------------------|
| 23/05/2014 09:45:35 | <INSTALLREQ>    | Installation is successful for 10000000             |
| 23/05/2014 09:45:35 | <INSTALLREQ>    | Installation request message received from 20100002 |
| 23/05/2014 09:45:35 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:36 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:37 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:38 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:40 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:41 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:42 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:43 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:45 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:46 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:47 | <INSTALLPARAMS> | Installation Parameters sent to 20100002            |
| 23/05/2014 09:45:49 | <RSSI>          | RSSI Response sent                                  |
| 23/05/2014 09:45:49 | <RSSI>          | RSSI Test Message 1 Received with RSSI:-17.5 dBm    |
| 23/05/2014 09:45:50 | <RSSI>          | RSSI Response sent.                                 |
| 23/05/2014 09:45:51 | <RSSI>          | RSSI Test Message 2 Received with RSSI:-19 dBm      |

The sequence of communications for EMT Installation is as follows:

| No. | Time     | Direction | Description                            | Message Type                        |
|-----|----------|-----------|----------------------------------------|-------------------------------------|
| 1   | 09:45:35 | Tx→Host   | Installation Request Received from Tx. | Parameter Request (0x10)            |
| 2   | 09:45:35 | Host→Tx   | Installation Parameters Sent to Tx.    | Field Installation Parameter (0x11) |
| 3   |          | Tx→Host   | Request Parameters Slave 1             | Modbus Parameter Request (0x12)     |
| 4   | 09:45:36 | Host→Tx   | Parameters for Modbus Slave 1          | Modbus Parameter (0x11)             |
| 5   |          | Tx→Host   | Request Parameters Slave 2             | Modbus Parameter Request (0x12)     |
| 6   | 09:45:37 | Host→Tx   | Parameters for Modbus Slave 2          | Modbus Parameter (0x11)             |
| 7   |          | Tx→Host   | Request Parameters Slave 3             | Modbus Parameter Request (0x12)     |
| 8   | 09:45:38 | Host→Tx   | Parameters for Modbus Slave 3          | Modbus Parameter (0x11)             |
| 9   |          | Tx→Host   | Request Parameters Slave 4             | Modbus Parameter Request (0x12)     |
| 10  | 09:45:40 | Host→Tx   | Parameters for Modbus Slave 4          | Modbus Parameter (0x11)             |
| 11  |          | Tx→Host   | Request Parameters Slave 5             | Modbus Parameter Request (0x12)     |
| 12  | 09:45:41 | Host→Tx   | Parameters for Modbus Slave 5          | Modbus Parameter (0x11)             |
| 13  |          | Tx→Host   | Request Parameters Slave 6             | Modbus Parameter Request (0x12)     |
| 14  | 09:45:42 | Host→Tx   | Parameters for Modbus Slave 6          | Modbus Parameter (0x11)             |
| 15  |          | Tx→Host   | Request Parameters Slave 7             | Modbus Parameter Request (0x12)     |
| 16  | 09:45:43 | Host→Tx   | Parameters for Modbus Slave 7          | Modbus Parameter (0x11)             |
| 17  |          | Tx→Host   | Request Parameters Slave 8             | Modbus Parameter Request (0x12)     |
| 18  | 09:45:45 | Host→Tx   | Parameters for Modbus Slave 8          | Modbus Parameter (0x11)             |
| 19  |          | Tx→Host   | Request Parameters Slave 9             | Modbus Parameter Request (0x12)     |
| 20  | 09:45:46 | Host→Tx   | Parameters for Modbus Slave 9          | Modbus Parameter (0x11)             |
| 21  |          | Tx→Host   | Request Parameters Slave 10            | Modbus Parameter Request (0x12)     |
| 22  | 09:45:47 | Host→Tx   | Parameters for Modbus Slave 10         | Modbus Parameter (0x11)             |

## Enless Data Transmission Message Format

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|    |          |         |                                   |                                |
|----|----------|---------|-----------------------------------|--------------------------------|
| 23 | 09:45:49 | Tx→Host | Acknowledgement Received From Tx. | Installation ACK (0x12)        |
| 24 | 09:45:49 | Host→Tx | RSSI response Sent to Tx.         | RSSI Response (0x15)           |
| 25 | 09:45:49 | Tx→Host | RSSI 1 Message Received From Tx.  | Field Installation RSSI (0x14) |
| 26 | 09:45:50 | Host→Tx | RSSI 1 response Sent to Tx.       | RSSI Response (0x15)           |
| 27 | 09:45:51 | Tx→Host | RSSI 2 Message Received From Tx.  | Field Installation RSSI (0x14) |
| 28 | 09:45:51 | Host→Tx | RSSI 2 response Sent to Tx.       | RSSI Response (0x15)           |
| 29 | 09:45:52 | Tx→Host | RSSI 3 Message Received From Tx.  | Field Installation RSSI (0x14) |
| 30 | 09:45:52 | Host→Tx | RSSI 3 response Sent to Tx.       | RSSI Response (0x15)           |
| 31 | 09:45:53 | Tx→Host | RSSI 4 Message Received From Tx.  | Field Installation RSSI (0x14) |
| 22 | 09:45:53 | Host→Tx | RSSI 4 response Sent to Tx.       | RSSI Response (0x15)           |
| 33 | 09:45:54 | Tx→Host | Installation Successful           | Installation Successful (0x16) |
| 44 |          | Tx→Host | Installation Successful           | Installation Successful (0x16) |

A Complete Installation Packet Sequence for the EMT (over-the-air) can be seen below (with the packet types highlighted in Red:

RX: Messages Received by Host from Enless Receiver

TX: Messages sent Out from Host to Enless Receiver

RX: \$68 \$15 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$02 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$10** \$05 \$29 \$16

TX: \$20 \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$05 \$0B \$20 \$10 \$00 \$02 \$00 \$05 \$00 \$01 \$00 \$40 \$20 \$00 \$00 \$03 \$E8 \$00 \$00 \$03 \$E8 \$01

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$03 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$12** \$05 \$00 \$44 \$16

TX: \$1C \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$05 \$0B \$20 \$10 \$00 \$02 \$01 \$05 \$00 \$01 \$00 \$40 \$20 \$65 \$8A \$03 \$E8 \$05

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$04 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$12** \$05 \$01 \$2A \$16

TX: \$1C \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$05 \$0B \$20 \$10 \$00 \$02 \$02 \$05 \$00 \$01 \$00 \$40 \$20 \$66 \$8A \$07 \$D0 \$05

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$05 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$12** \$05 \$02 \$23 \$16

TX: \$1C \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$05 \$0B \$20 \$10 \$00 \$02 \$03 \$05 \$00 \$01 \$00 \$40 \$20 \$67 \$8A \$0B \$B8 \$05

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$06 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$12** \$05 \$03 \$22 \$16

TX: \$1C \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$05 \$0B \$20 \$10 \$00 \$02 \$04 \$05 \$00 \$01 \$00 \$40 \$20 \$68 \$8A \$0F \$A0 \$05

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$07 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$12** \$05 \$04 \$22 \$16

TX: \$1C \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$05 \$0B \$20 \$10 \$00 \$02 \$05 \$05 \$00 \$01 \$00 \$40 \$20 \$69 \$8A \$13 \$88 \$05

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$08 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$12** \$05 \$05 \$22 \$16

TX: \$1C \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$05 \$0B \$20 \$10 \$00 \$02 \$06 \$05 \$00 \$01 \$00 \$40 \$20 \$6A \$8A \$17 \$70 \$05

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$09 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$12** \$05 \$06 \$22 \$16

TX: \$1C \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$05 \$0B \$20 \$10 \$00 \$02 \$07 \$05 \$00 \$01 \$00 \$40 \$20 \$6B \$8A \$1B \$58 \$05

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$0A \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$12** \$05 \$07 \$23 \$16

TX: \$1C \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$05 \$0B \$20 \$10 \$00 \$02 \$08 \$05 \$00 \$01 \$00 \$40 \$20 \$6C \$8A \$1F \$40 \$05

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$0B \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$12** \$05 \$08 \$22 \$16

TX: \$1C \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$05 \$0B \$20 \$10 \$00 \$02 \$09 \$05 \$00 \$01 \$00 \$40 \$20 \$6D \$8A \$23 \$28 \$05

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$0C \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$12** \$05 \$09 \$22 \$16

TX: \$1C \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$05 \$0B \$20 \$10 \$00 \$02 \$0A \$05 \$00 \$01 \$00 \$40 \$20 \$6E \$8A \$27 \$10 \$05

## Enless Data Transmission Message Format

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RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$0D \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$12 \$05 \$0A \$22 \$16  
TX: \$11 \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$15 \$05 \$01 \$20 \$10 \$00 \$02 \$18

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$0E \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$14 \$05 \$02 \$23 \$16  
TX: \$11 \$7A \$76 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$15 \$05 \$02 \$20 \$10 \$00 \$02 \$18

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$0F \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$14 \$05 \$03 \$23 \$16  
TX: \$11 \$7A \$77 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$15 \$05 \$03 \$20 \$10 \$00 \$02 \$1B

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$10 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$14 \$05 \$04 \$22 \$16  
TX: \$11 \$7A \$78 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$15 \$05 \$04 \$20 \$10 \$00 \$02 \$19

RX: \$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$11 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$14 \$05 \$05 \$23 \$16  
TX: \$11 \$7A \$79 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$15 \$05 \$05 \$20 \$10 \$00 \$02 \$19

RX: \$68 \$15 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$12 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$16 \$05 \$23 \$16  
RX: \$68 \$15 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$13 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$16 \$05 \$23 \$16

**The message structure is exactly the same as 'over-the-air' except for the messages sent back from the host to the receiver.**

**These messages start from the CI field and do not require the Radiocrafts WMBUS bytes to be transmitted. The first byte of the packet sent back to the receiver is the length field.**

## 4.1. Installation Request

| Message ID | Device Type |
|------------|-------------|
| 0x10       | 0x01 – 0xFF |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:
  1. 0x01 - Temperature Transmitter
  2. 0x02 is the Temperature and Humidity Transmitter
  3. 0x03 is the PT100 Temperature Device
  4. 0x04 is Pulse Transmitter Device
  5. 0x05 Energy Meter Transmitter Device
  6. 0x06 Repeater
  7. 0x07 Modbus Receiver

A typical packet (over-the-air) breakdown is shown below:

| INSTALLATION REQUEST |                                                  |              |                  |             |            |            |       |
|----------------------|--------------------------------------------------|--------------|------------------|-------------|------------|------------|-------|
| <b>Data Frame</b>    | 681544AE0C9900901001077A010000002F2F0F7F10044D16 |              |                  |             |            |            |       |
|                      | <b>Start</b>                                     | <b>Count</b> | <b>HEX VALUE</b> | <b>DEC</b>  |            |            |       |
| Start Byte           | 1                                                | 2            | 68               |             |            |            |       |
| Length Byte          | 3                                                | 2            | 15               | 21          |            |            |       |
| C Field              | 5                                                | 2            | 44               |             |            |            |       |
| M Field              | 7                                                | 4            | AE0C             |             |            |            |       |
| A Field              | 11                                               | 12           | 990090100107     |             |            |            |       |
| CI Control           | 23                                               | 2            | 7A               |             |            |            |       |
| Access Number        | 25                                               | 2            | 01               |             |            |            |       |
| Status               | 27                                               | 2            | 00               |             |            |            |       |
| Config Word          | 29                                               | 4            | 0000             |             |            |            |       |
| AES Verify           | 33                                               | 4            | 2F2F             |             |            |            |       |
| DIF                  | 37                                               | 2            | 0F               |             |            |            |       |
| VIF                  | 39                                               | 2            | 7F               |             |            |            |       |
|                      |                                                  |              |                  |             | <b>HEX</b> | <b>DEC</b> |       |
| Data                 | 41                                               | 4            | 1004             | <b>ID</b>   | 10         | 16         |       |
|                      |                                                  |              |                  | <b>Type</b> | 04         | 4          |       |
|                      |                                                  |              |                  |             |            |            |       |
| RSSI                 | 45                                               | 2            | 4D               |             | 4D         | 77         | -38.5 |
| Stop                 | 47                                               | 2            | 16               |             |            |            |       |

Received Packet (same as over the air)

\$68 \$15 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$01 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$10 \$03 \$22 \$16

## 4.2. *TX\_TEMP Installation*

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x01 - Temperature Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.



## Enless Data Transmission Message Format

A typical packet breakdown is shown below:

| TEMPERATURE INSTALLATION PACKET |                                                                        |       |                            |               |          |       |
|---------------------------------|------------------------------------------------------------------------|-------|----------------------------|---------------|----------|-------|
| Data Frame                      | 68204424487856341201077A1E0000002F2F0F7F110101105000240500010040204616 |       |                            |               |          |       |
|                                 | Start                                                                  | Count | HEX VALUE                  | DEC           |          |       |
| Start Byte                      | 1                                                                      | 2     | 68                         |               |          |       |
| Length Byte                     | 3                                                                      | 2     | 20                         | 32            |          |       |
| C Field                         | 5                                                                      | 2     | 44                         |               |          |       |
| M Field                         | 7                                                                      | 4     | 2448                       |               |          |       |
| A Field                         | 11                                                                     | 12    | 785634120107               |               |          |       |
| CI Control                      | 23                                                                     | 2     | 7A                         |               |          |       |
| Access Number                   | 25                                                                     | 2     | 1E                         |               |          |       |
| Status                          | 27                                                                     | 2     | 00                         |               |          |       |
| Config Word                     | 29                                                                     | 4     | 0000                       |               |          |       |
| AES Verify                      | 33                                                                     | 4     | 2F2F                       |               |          |       |
| DIF                             | 37                                                                     | 2     | 0F                         |               |          |       |
| VIF                             | 39                                                                     | 2     | 7F                         |               |          |       |
|                                 |                                                                        |       |                            | HEX           | DEC      |       |
| Data                            | 41                                                                     | 26    | 11010110500024050001004020 | ID            | 11       | 17    |
|                                 |                                                                        |       |                            | Device Type   | 01       | 1     |
|                                 |                                                                        |       |                            | Frame Counter | 01       | 1     |
|                                 |                                                                        |       |                            | TX ID         | 10500024 |       |
|                                 |                                                                        |       |                            | Tx Interval   | 05       | 5     |
|                                 |                                                                        |       |                            | Re-Try Count  | 00       | 0     |
|                                 |                                                                        |       |                            | Receiver ID   | 01004020 |       |
| RSSI                            | 67                                                                     | 2     | 46                         | 46            | 70       | -35.0 |
| Stop                            | 69                                                                     | 2     | 16                         |               |          |       |

Below are typical request and response packets for this message.

Received Packet (same as over the air) :

\$68 \$15 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$01 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$10 \$03 \$22 \$16

Host will Respond with the following:

\$16 \$7A \$5D \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$11 \$01 \$01 \$10 \$50 \$00 \$18 \$05 \$00 \$01 \$00 \$40 \$20

**Note the response from the host starts from the CI field to the end of the data field and the first byte of the packet is the length field.**

### 4.3. *TX\_TEMP\_HUM Installation*

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x02 - Temperature and Humidity Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.

## Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| TEMPERATURE HUMIDITY TRANSMITTER INSTALLATION PACKET |                                                                        |       |                            |               |          |       |
|------------------------------------------------------|------------------------------------------------------------------------|-------|----------------------------|---------------|----------|-------|
| Data Frame                                           | 68204424487856341201077A1E0000002F2F0F7F110201105000110300010040204616 |       |                            |               |          |       |
|                                                      | Start                                                                  | Count | HEX VALUE                  | DEC           |          |       |
| Start Byte                                           | 1                                                                      | 2     | 68                         |               |          |       |
| Length Byte                                          | 3                                                                      | 2     | 20                         | 32            |          |       |
| C Field                                              | 5                                                                      | 2     | 44                         |               |          |       |
| M Field                                              | 7                                                                      | 4     | 2448                       |               |          |       |
| A Field                                              | 11                                                                     | 12    | 785634120107               |               |          |       |
| CI Control                                           | 23                                                                     | 2     | 7A                         |               |          |       |
| Access Number                                        | 25                                                                     | 2     | 1E                         |               |          |       |
| Status                                               | 27                                                                     | 2     | 00                         |               |          |       |
| Config Word                                          | 29                                                                     | 4     | 0000                       |               |          |       |
| AES Verify                                           | 33                                                                     | 4     | 2F2F                       |               |          |       |
| DIF                                                  | 37                                                                     | 2     | 0F                         |               |          |       |
| VIF                                                  | 39                                                                     | 2     | 7F                         |               |          |       |
|                                                      |                                                                        |       |                            |               | HEX      | DEC   |
| Data                                                 | 41                                                                     | 26    | 11020110500011030001004020 | ID            | 11       | 17    |
|                                                      |                                                                        |       |                            | Device Type   | 02       | 2     |
|                                                      |                                                                        |       |                            | Frame Counter | 01       | 1     |
|                                                      |                                                                        |       |                            | TX ID         | 10500011 |       |
|                                                      |                                                                        |       |                            | Tx Interval   | 03       | 3     |
|                                                      |                                                                        |       |                            | Re-Try Count  | 00       | 0     |
|                                                      |                                                                        |       |                            | Receiver ID   | 01004020 |       |
| RSSI                                                 | 67                                                                     | 2     | 46                         |               | 46       | 70    |
| Stop                                                 | 69                                                                     | 2     | 16                         |               |          | -35.0 |

Below are typical request and response packets for this message.

Received Packet (same as over the air) :

\$68 \$15 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$01 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$10 \$03 \$22 \$16

Host will Respond with the following:

\$16 \$7A \$5D \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$11 \$02 \$01 \$10 \$50 \$00 \$18 \$05 \$00 \$01 \$00 \$40 \$20

**Note the response from the host starts from the CI field to the end of the data field and the first byte of the packet is the length field.**

#### 4.4. *TX\_TEMP\_PT100 Installation*

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x03 - PT100 Temperature
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min.
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.

## Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| PT100 TRANSMITTER INSTALLATION PACKET |                                                                        |       |                            |               |          |          |
|---------------------------------------|------------------------------------------------------------------------|-------|----------------------------|---------------|----------|----------|
| Data Frame                            | 68204424487856341201077A1E0000002F2F0F7F110301105000860500010040204616 |       |                            |               |          |          |
|                                       | Start                                                                  | Count | HEX VALUE                  | DEC           |          |          |
| Start Byte                            | 1                                                                      | 2     | 68                         |               |          |          |
| Length Byte                           | 3                                                                      | 2     | 20                         | 32            |          |          |
| C Field                               | 5                                                                      | 2     | 44                         |               |          |          |
| M Field                               | 7                                                                      | 4     | 2448                       |               |          |          |
| A Field                               | 11                                                                     | 12    | 785634120107               |               |          |          |
| CI Control                            | 23                                                                     | 2     | 7A                         |               |          |          |
| Access Number                         | 25                                                                     | 2     | 1E                         |               |          |          |
| Status                                | 27                                                                     | 2     | 00                         |               |          |          |
| Config Word                           | 29                                                                     | 4     | 0000                       |               |          |          |
| AES Verify                            | 33                                                                     | 4     | 2F2F                       |               |          |          |
| DIF                                   | 37                                                                     | 2     | 0F                         |               |          |          |
| VIF                                   | 39                                                                     | 2     | 7F                         |               |          |          |
|                                       |                                                                        |       |                            |               | HEX      | DEC      |
| Data                                  | 41                                                                     | 26    | 11030110500086050001004020 | ID            | 11       | 17       |
|                                       |                                                                        |       |                            | Device Type   | 03       | 3        |
|                                       |                                                                        |       |                            | Frame Counter | 01       | 1        |
|                                       |                                                                        |       |                            | TX ID         | 10500086 | 10900099 |
|                                       |                                                                        |       |                            | Tx Interval   | 05       | 5        |
|                                       |                                                                        |       |                            | Re-Try Count  | 00       | 0        |
|                                       |                                                                        |       |                            | Receiver ID   | 01004020 |          |
| RSSI                                  | 67                                                                     | 2     | 46                         |               | 46       | 70       |
| Stop                                  | 69                                                                     | 2     | 16                         |               |          | -35.0    |

Below are typical request and response packets for this message.

Received Packet (same as over the air) :

\$68 \$15 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$01 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$10 \$03 \$22 \$16

Host will Respond with the following:

\$16 \$7A \$5D \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$11 \$03 \$01 \$10 \$50 \$00 \$18 \$05 \$00 \$01 \$00 \$40 \$20

**Note the response from the host starts from the CI field to the end of the data field and the first byte of the packet is the length field.**

### 4.5. *TX\_PULSE Installation*

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x04 is Pulse Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below:

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 | Pulse1 Count D0 | Pulse1 Count D1 | Pulse1 Count D2 | Pulse1 Count D3 | Pulse2 Count D0 | Pulse2 Count D1 | Pulse Count D2 | Pulse1 Count D3 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|-----------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                | 0x00            | 0x00            | 0x00            | 0x00            | 0x00            | 0x00            | 0x00           | 0x0             |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.
- **Pulse Count 1 and Pulse Count2** – it will be 4 byte initial count for both pulse count 1 and pulse count 2 user can set initial count from 0x00000000 to less than 0xFFFFFFFF.

## Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| PULSE TRANSMITTER INSTALLATION PACKET |                                                                                        |       |                                            |                       |          |       |
|---------------------------------------|----------------------------------------------------------------------------------------|-------|--------------------------------------------|-----------------------|----------|-------|
| Data Frame                            | 68284424487856341201077A0A0000002F2F0F7F110401109000990500010040200000000A000000643516 |       |                                            |                       |          |       |
|                                       | Start                                                                                  | Count | HEX VALUE                                  | DEC                   |          |       |
| Start Byte                            | 1                                                                                      | 2     | 68                                         |                       |          |       |
| Length Byte                           | 3                                                                                      | 2     | 28                                         | 40                    |          |       |
| C Field                               | 5                                                                                      | 2     | 44                                         |                       |          |       |
| M Field                               | 7                                                                                      | 4     | 2448                                       |                       |          |       |
| A Field                               | 11                                                                                     | 12    | 785634120107                               |                       |          |       |
| CI Control                            | 23                                                                                     | 2     | 7A                                         |                       |          |       |
| Access Number                         | 25                                                                                     | 2     | 0A                                         |                       |          |       |
| Status                                | 27                                                                                     | 2     | 00                                         |                       |          |       |
| Config Word                           | 29                                                                                     | 4     | 0000                                       |                       |          |       |
| AES Verify                            | 33                                                                                     | 4     | 2F2F                                       |                       |          |       |
| DIF                                   | 37                                                                                     | 2     | 0F                                         |                       |          |       |
| VIF                                   | 39                                                                                     | 2     | 7F                                         |                       |          |       |
|                                       |                                                                                        |       |                                            | HEX                   | DEC      |       |
| Data                                  | 41                                                                                     | 42    | 110401109000990500010040200000000A00000064 | ID                    | 11       | 17    |
|                                       |                                                                                        |       |                                            | Device Type           | 04       | 4     |
|                                       |                                                                                        |       |                                            | Frame Counter         | 01       | 1     |
|                                       |                                                                                        |       |                                            | TX ID                 | 10900099 |       |
|                                       |                                                                                        |       |                                            | Tx Interval           | 05       | 5     |
|                                       |                                                                                        |       |                                            | Re-Try Count          | 00       | 0     |
|                                       |                                                                                        |       |                                            | Receiver ID           | 01004020 |       |
|                                       |                                                                                        |       |                                            | Initial Pulse Count 1 | 0000000A | 10    |
|                                       |                                                                                        |       |                                            | InitialPulse Count 2  | 00000064 | 100   |
| RSSI                                  | 83                                                                                     | 2     | 35                                         |                       | 35       | 53    |
| Stop                                  | 85                                                                                     | 2     | 16                                         |                       |          | -26.5 |

Below are typical request and response packets for this message.

Received Packet (same as over the air)

\$68 \$15 \$44 \$AE \$0C \$99 \$00 \$90 \$10 \$01 \$07 \$7A \$01 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$10 \$04 \$41 \$16

Host will Respond with the following:

\$1E \$7A \$49 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$11 \$04 \$01 \$10 \$90 \$00 \$99 \$05 \$00 \$01 \$00 \$40 \$20 \$00 \$00 \$00 \$0A \$00 \$00 \$00 \$64

**Note the response from the host starts from the CI field to the end of the data field and the first byte of the packet is the length field.**

## 4.6 TX\_CONTACT Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where: 0x23 is Contact Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below:

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.



Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| CONTACT INSTALLATION PACKET |                                                                        |       |                            |  |               |       |     |          |       |
|-----------------------------|------------------------------------------------------------------------|-------|----------------------------|--|---------------|-------|-----|----------|-------|
| Data Frame                  | 682044AE0C0100002001077A000000002F2F0F7F112301111111130100000000003C16 |       |                            |  |               |       |     |          |       |
|                             | Start                                                                  | Count | HEX VALUE                  |  | DEC           |       |     |          |       |
| Start Byte                  | 1                                                                      | 2     | 68                         |  |               |       |     |          |       |
| Length Byte                 | 3                                                                      | 2     | 20                         |  | 32            |       |     |          |       |
| C Field                     | 5                                                                      | 2     | 44                         |  |               |       |     |          |       |
| M Field                     | 7                                                                      | 4     | AE0C                       |  |               |       |     |          |       |
| A Field                     | 11                                                                     | 12    | 010000200107               |  |               |       |     |          |       |
| CI Control                  | 23                                                                     | 2     | 7A                         |  |               |       |     |          |       |
| Access Number               | 25                                                                     | 2     | 00                         |  |               |       |     |          |       |
| Status                      | 27                                                                     | 2     | 00                         |  |               |       |     |          |       |
| Config Word                 | 29                                                                     | 4     | 0000                       |  |               |       |     |          |       |
| AES Verify                  | 33                                                                     | 4     | 2F2F                       |  |               |       |     |          |       |
| DIF                         | 37                                                                     | 2     | 0F                         |  |               |       |     |          |       |
| VIF                         | 39                                                                     | 2     | 7F                         |  |               |       |     |          |       |
|                             |                                                                        |       |                            |  | Start         | Count | HEX | DEC      |       |
| Data                        | 41                                                                     | 26    | 11230111111113010000000000 |  | ID            | 41    | 2   | 11       | 17    |
|                             |                                                                        |       |                            |  | Device Type   | 43    | 2   | 23       | 35    |
|                             |                                                                        |       |                            |  | Frame Counter | 45    | 2   | 01       | 1     |
|                             |                                                                        |       |                            |  | TX ID         | 47    | 8   | 11111113 |       |
|                             |                                                                        |       |                            |  | Tx Interval   | 55    | 2   | 01       | 1     |
|                             |                                                                        |       |                            |  | Re-Try Count  | 57    | 2   | 00       | 0     |
|                             |                                                                        |       |                            |  | Receiver ID   | 59    | 8   | 00000000 |       |
| RSSI                        | 67                                                                     | 2     | 3C                         |  |               |       |     |          |       |
| Stop                        | 69                                                                     | 2     | 16                         |  |               |       |     |          |       |
|                             |                                                                        |       |                            |  |               |       |     | 3C       | 60    |
|                             |                                                                        |       |                            |  |               |       |     |          | -30.0 |

Received Packet (same as over the air)

[68, 15, 44, AE, 0C, 13, 11, 11, 11, 01, 07, 7A, 01, 00, 00, 00, 2F, 2F, 0F, 7F, 10, 23, 5B, 16]

Host will Respond with the following:

16 7A 00 00 00 00 2F 2F 0F 7F 11 23 01 11 11 11 13 01 00 00 00 00 00

Note the response from the host starts from the CI field to the end of the data field and the first byte of the packet is the length field.

## 4.7 TX\_CO2 Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x24 is CO2 Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below:

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.

## Enless Data Transmission Message Format

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A typical packet (over-the-air) breakdown is shown below:

| CO2 INSTALLATION PACKET |                                                                        |       |                            |     |               |       |     |          |       |
|-------------------------|------------------------------------------------------------------------|-------|----------------------------|-----|---------------|-------|-----|----------|-------|
| Data Frame              | 682044AE0C0100002001077A000000002F2F0F7F112401104000010101000000003A16 |       |                            |     |               |       |     |          |       |
|                         | Start                                                                  | Count | HEX VALUE                  | DEC |               |       |     |          |       |
| Start Byte              | 1                                                                      | 2     | 68                         |     |               |       |     |          |       |
| Length Byte             | 3                                                                      | 2     | 20                         | 32  |               |       |     |          |       |
| C Field                 | 5                                                                      | 2     | 44                         |     |               |       |     |          |       |
| M Field                 | 7                                                                      | 4     | AE0C                       |     |               |       |     |          |       |
| A Field                 | 11                                                                     | 12    | 010000200107               |     |               |       |     |          |       |
| CI Control              | 23                                                                     | 2     | 7A                         |     |               |       |     |          |       |
| Access Number           | 25                                                                     | 2     | 00                         |     |               |       |     |          |       |
| Status                  | 27                                                                     | 2     | 00                         |     |               |       |     |          |       |
| Config Word             | 29                                                                     | 4     | 0000                       |     |               |       |     |          |       |
| AES Verify              | 33                                                                     | 4     | 2F2F                       |     |               |       |     |          |       |
| DIF                     | 37                                                                     | 2     | 0F                         |     |               |       |     |          |       |
| VIF                     | 39                                                                     | 2     | 7F                         |     |               |       |     |          |       |
|                         |                                                                        |       |                            |     | Start         | Count | HEX | DEC      |       |
| Data                    | 41                                                                     | 26    | 11240110400001010100000000 |     | ID            | 41    | 2   | 11       | 17    |
|                         |                                                                        |       |                            |     | Device Type   | 43    | 2   | 24       | 36    |
|                         |                                                                        |       |                            |     | Frame Counter | 45    | 2   | 01       | 1     |
|                         |                                                                        |       |                            |     | TX ID         | 47    | 8   | 10400001 |       |
|                         |                                                                        |       |                            |     | Tx Interval   | 55    | 2   | 01       | 1     |
|                         |                                                                        |       |                            |     | Re-Try Count  | 57    | 2   | 01       | 1     |
|                         |                                                                        |       |                            |     | Receiver ID   | 59    | 8   | 00000000 |       |
| RSSI                    | 67                                                                     | 2     | 3A                         |     |               |       |     | 3A       | 58    |
| Stop                    | 69                                                                     | 2     | 16                         |     |               |       |     |          | -29.0 |

Received Packet (same as over the air)

[68, 15, 44, AE, 0C, 01, 00, 40, 10, 01, 07, 7A, 01, 00, 00, 00, 2F, 2F, 0F, 7F, 10, 24, 5F, 16]

Host will Respond with the following:

16 7A 00 00 00 00 2F 2F 0F 7F 11 24 01 10 40 00 01 01 01 00 00 00 00

**Note the response from the host starts from the CI field to the end of the data field and the first byte of the packet is the length field.**

## 4.8 TX\_4to20 Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x25 is 4-20mA Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below:

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 | Sensor Power Time D0 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|----------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                | 0x11                 |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.
- **Sensor Power Time (D0)** – one byte and denotes how many seconds the sensor will be powered for before transmission.

Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| 4-20mA INSTALLATION PACKET |                                                                         |       |                             |  |               |    |   |          |       |
|----------------------------|-------------------------------------------------------------------------|-------|-----------------------------|--|---------------|----|---|----------|-------|
| Data Frame                 | 682144AE0C0100002001077A000000002F2F0F7F1125011111112010100000000023A16 |       |                             |  |               |    |   |          |       |
|                            | Start                                                                   | Count | HEX VALUE                   |  | DEC           |    |   |          |       |
| Start Byte                 | 1                                                                       | 2     | 68                          |  |               |    |   |          |       |
| Length Byte                | 3                                                                       | 2     | 21                          |  | 33            |    |   |          |       |
| C Field                    | 5                                                                       | 2     | 44                          |  |               |    |   |          |       |
| M Field                    | 7                                                                       | 4     | AE0C                        |  |               |    |   |          |       |
| A Field                    | 11                                                                      | 12    | 010000200107                |  |               |    |   |          |       |
| CI Control                 | 23                                                                      | 2     | 7A                          |  |               |    |   |          |       |
| Access Number              | 25                                                                      | 2     | 00                          |  |               |    |   |          |       |
| Status                     | 27                                                                      | 2     | 00                          |  |               |    |   |          |       |
| Config Word                | 29                                                                      | 4     | 0000                        |  |               |    |   |          |       |
| AES Verify                 | 33                                                                      | 4     | 2F2F                        |  |               |    |   |          |       |
| DIF                        | 37                                                                      | 2     | 0F                          |  |               |    |   |          |       |
| VIF                        | 39                                                                      | 2     | 7F                          |  |               |    |   |          |       |
| Data                       | 41                                                                      | 28    | 112501111111201010000000002 |  | ID            | 41 | 2 | 11       | 17    |
|                            |                                                                         |       |                             |  | Device Type   | 43 | 2 | 25       | 37    |
|                            |                                                                         |       |                             |  | Frame Counter | 45 | 2 | 01       | 1     |
|                            |                                                                         |       |                             |  | TX ID         | 47 | 8 | 11111112 |       |
|                            |                                                                         |       |                             |  | Tx Interval   | 55 | 2 | 01       | 1     |
|                            |                                                                         |       |                             |  | Re-Try Count  | 57 | 2 | 01       | 1     |
|                            |                                                                         |       |                             |  | Receiver ID   | 59 | 8 | 00000000 |       |
|                            |                                                                         |       |                             |  | Wake Period   | 67 | 2 | 02       | 2     |
| RSSI                       | 69                                                                      | 2     | 3A                          |  |               |    |   | 3A       | 58    |
| Stop                       | 71                                                                      | 2     | 16                          |  |               |    |   |          | -29.0 |

Received Packet (same as over the air)

68, 15, 44, AE, 0C, 12, 11, 11, 11, 01, 07, 7A, 01, 00, 00, 00, 2F, 2F, 0F, 7F, 10, 25, 5B, 16

Host will Respond with the following:

17 7A 00 00 00 00 2F 2F 0F 7F 11 25 01 11 11 11 12 01 01 00 00 00 00 02

Note the response from the host starts from the CI field to the end of the data field and the first byte of the packet is the length field.

### 4.9 TX\_0to5 Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x26 is 0-5V Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below:

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 | Sensor Power Time D0 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|----------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                | 0x11                 |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.
- **Sensor Power Time (D0)** – one byte and denotes how many seconds the sensor will be powered for before transmission.

## Enless Data Transmission Message Format

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A typical packet (over-the-air) breakdown is shown below:

| 0-5V INSTALLATION PACKET |                                                                         |       |                             |     |               |    |   |          |       |
|--------------------------|-------------------------------------------------------------------------|-------|-----------------------------|-----|---------------|----|---|----------|-------|
| Data Frame               | 682144AE0C0100002001077A000000002F2F0F7F1126011111110010100000000043A16 |       |                             |     |               |    |   |          |       |
|                          | Start                                                                   | Count | HEX VALUE                   | DEC |               |    |   |          |       |
| Start Byte               | 1                                                                       | 2     | 68                          |     |               |    |   |          |       |
| Length Byte              | 3                                                                       | 2     | 21                          | 33  |               |    |   |          |       |
| C Field                  | 5                                                                       | 2     | 44                          |     |               |    |   |          |       |
| M Field                  | 7                                                                       | 4     | AEOC                        |     |               |    |   |          |       |
| A Field                  | 11                                                                      | 12    | 010000200107                |     |               |    |   |          |       |
| CI Control               | 23                                                                      | 2     | 7A                          |     |               |    |   |          |       |
| Access Number            | 25                                                                      | 2     | 00                          |     |               |    |   |          |       |
| Status                   | 27                                                                      | 2     | 00                          |     |               |    |   |          |       |
| Config Word              | 29                                                                      | 4     | 0000                        |     |               |    |   |          |       |
| AES Verify               | 33                                                                      | 4     | 2F2F                        |     |               |    |   |          |       |
| DIF                      | 37                                                                      | 2     | 0F                          |     |               |    |   |          |       |
| VIF                      | 39                                                                      | 2     | 7F                          |     |               |    |   |          |       |
| Data                     | 41                                                                      | 28    | 112601111111001010000000004 |     | ID            | 41 | 2 | 11       | 17    |
|                          |                                                                         |       |                             |     | Device Type   | 43 | 2 | 26       | 38    |
|                          |                                                                         |       |                             |     | Frame Counter | 45 | 2 | 01       | 1     |
|                          |                                                                         |       |                             |     | TX ID         | 47 | 8 | 11111110 |       |
|                          |                                                                         |       |                             |     | Tx Interval   | 55 | 2 | 01       | 1     |
|                          |                                                                         |       |                             |     | Re-Try Count  | 57 | 2 | 01       | 1     |
|                          |                                                                         |       |                             |     | Receiver ID   | 59 | 8 | 00000000 |       |
|                          |                                                                         |       |                             |     | Wake Period   | 67 | 2 | 04       | 4     |
| RSSI                     | 69                                                                      | 2     | 3A                          |     |               |    |   | 3A       | 58    |
| Stop                     | 71                                                                      | 2     | 16                          |     |               |    |   |          | -29.0 |

Received Packet (same as over the air)

68, 15, 44, AE, 0C, 10, 11, 11, 11, 01, 07, 7A, 01, 00, 00, 00, 2F, 2F, 0F, 7F, 10, 26, 71, 16

Host will Respond with the following:

17 7A 00 00 00 00 2F 2F 0F 7F 11 26 01 11 11 11 10 01 01 00 00 00 00 04

**Note the response from the host starts from the CI field to the end of the data field and the first byte of the packet is the length field.**

## 4.10 TX\_0to10 Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See Below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x27 is 0-10V Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below:

| Transmit Interval | Retry Count | Receiver Address D0 | Receiver Address D1 | Receiver Address D2 | Receiver Address D3 | Sensor Power Time D0 |
|-------------------|-------------|---------------------|---------------------|---------------------|---------------------|----------------------|
| 0x03 – 0xFF       | 0x00 – 0x01 | 0x44                | 0x33                | 0x22                | 0x11                | 0x11                 |

- **Transmit Interval** – It will be 0x01 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 1 to 255 minute. If set to 0x00, the transmitter will default to 5min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of Receiver address. In given example 0x11223344.
- **Sensor Power Time (D0)** – one byte and denotes how many seconds the sensor will be powered for before transmission.



## Enless Data Transmission Message Format

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A typical packet (over-the-air) breakdown is shown below:

| 0-10V INSTALLATION PACKET |                                                                         |       |                             |     |               |    |   |          |       |
|---------------------------|-------------------------------------------------------------------------|-------|-----------------------------|-----|---------------|----|---|----------|-------|
| Data Frame                | 682144AE0C0100002001077A000000002F2F0F7F1127011111111010100000000053916 |       |                             |     |               |    |   |          |       |
|                           | Start                                                                   | Count | HEX VALUE                   | DEC |               |    |   |          |       |
| Start Byte                | 1                                                                       | 2     | 68                          |     |               |    |   |          |       |
| Length Byte               | 3                                                                       | 2     | 21                          | 33  |               |    |   |          |       |
| C Field                   | 5                                                                       | 2     | 44                          |     |               |    |   |          |       |
| M Field                   | 7                                                                       | 4     | AE0C                        |     |               |    |   |          |       |
| A Field                   | 11                                                                      | 12    | 010000200107                |     |               |    |   |          |       |
| CI Control                | 23                                                                      | 2     | 7A                          |     |               |    |   |          |       |
| Access Number             | 25                                                                      | 2     | 00                          |     |               |    |   |          |       |
| Status                    | 27                                                                      | 2     | 00                          |     |               |    |   |          |       |
| Config Word               | 29                                                                      | 4     | 0000                        |     |               |    |   |          |       |
| AES Verify                | 33                                                                      | 4     | 2F2F                        |     |               |    |   |          |       |
| DIF                       | 37                                                                      | 2     | 0F                          |     |               |    |   |          |       |
| VIF                       | 39                                                                      | 2     | 7F                          |     |               |    |   |          |       |
| Data                      | 41                                                                      | 28    | 112701111111101010000000005 |     | ID            | 41 | 2 | 11       | 17    |
|                           |                                                                         |       |                             |     | Device Type   | 43 | 2 | 27       | 39    |
|                           |                                                                         |       |                             |     | Frame Counter | 45 | 2 | 01       | 1     |
|                           |                                                                         |       |                             |     | TX ID         | 47 | 8 | 11111111 |       |
|                           |                                                                         |       |                             |     | Tx Interval   | 55 | 2 | 01       | 1     |
|                           |                                                                         |       |                             |     | Re-Try Count  | 57 | 2 | 01       | 1     |
|                           |                                                                         |       |                             |     | Receiver ID   | 59 | 8 | 00000000 |       |
|                           |                                                                         |       |                             |     | Wake Period   | 67 | 2 | 05       | 5     |
| RSSI                      | 69                                                                      | 2     | 39                          |     |               |    |   | 39       | 57    |
| Stop                      | 71                                                                      | 2     | 16                          |     |               |    |   |          | -28.5 |

Received Packet (same as over the air)

68, 15, 44, AE, 0C, 11, 11, 11, 11, 01, 07, 7A, 01, 00, 00, 00, 2F, 2F, 0F, 7F, 10, 27, 5B, 16

Host will Respond with the following:

17 7A 00 00 00 00 2F 2F 0F 7F 11 27 01 11 11 11 11 01 01 00 00 00 00 05

**Note the response from the host starts from the CI field to the end of the data field and the first byte of the packet is the length field.**

## 4.11 TX\_ENERGY Installation

| Message ID | Device Type | Msg Frame Counter | Destination Address D0 | Destination Address D1 | Destination Address D2 | Destination Address D3 | Data      |
|------------|-------------|-------------------|------------------------|------------------------|------------------------|------------------------|-----------|
| 0x11       | 0x01 – 0xFF | 0x01 - 0x0B       | 0x44                   | 0x33                   | 0x22                   | 0x11                   | See below |

- **Message ID** – It will be always 0x10 is Field Installation Request Message ID.
- **Device Type** – It will be 0x01 – 0xFF where:  
0x05 Energy Meter Transmitter
- **Message Frame Counter** – It will be 0x01 to 0x0B count for referring installation parameter sent from Receiver and device based frame counter will sent from Receiver.  
For example Temperature Transmitter Device it will be 0x01 which Receiver sent single message to transmitter. And for Energy Meter there 11 message need to send from Receiver to install 10 slave meter and 1 pulse counter message.
- **Destination Address** – it will 4 bytes of data which is transmitter device address sent from Receiver. In the given example is 0x11223344.
- **Data** – It will be depends on type device, see below table

### 4.11.1 TX\_ENERGY Pulse Installation

| Message Number | Transmit Interval | Retry Count | Recvr Address D0 | Recvr Address D1 | Recvr Address D2 | Recvr Address D3 | Pulse1 Count D0 | Pulse1 Count D1 | Pulse1 Count D2 | Pulse1 Count D3 | Pulse2 Count D0 | Pulse2 Count D1 | Pulse Count D2 | Pulse1 Count D3 |
|----------------|-------------------|-------------|------------------|------------------|------------------|------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|-----------------|
| 0x00           | 0x03 – 0xFF       | 0x00 – 0x01 | 0x44             | 0x33             | 0x22             | 0x11             | 0x00            | 0x00            | 0x00            | 0x00            | 0x00            | 0x00            | 0x00           | 0x0             |

- **Message Number** – it will be always 0x00 for Energy meter pulse counter device.
- **Transmit Interval** – It will be 0x03 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 3 to 255 minute. If set to below 0x03, the transmitter will default to 3min
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of transmitter address. In given example 0x11223344.
- **Pulse Count 1 and Pulse Count2** – it will be 4 byte initial count for both pulse count 1 and pulse count 2 user can set initial count from 0x00000000 to less than 0xFFFFFFFF.

## Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| EMT PULSE TRANSMITTER INSTALLATION PACKET |                                                                                            |       |                                                |                       |          |       |
|-------------------------------------------|--------------------------------------------------------------------------------------------|-------|------------------------------------------------|-----------------------|----------|-------|
| Data Frame                                | 682A4424487856341201077A280000002F2F0F7F11050B2010000200050001004020000003E8000003E8013B16 |       |                                                |                       |          |       |
|                                           | Start                                                                                      | Count | HEX VALUE                                      | DEC                   |          |       |
| Start Byte                                | 1                                                                                          | 2     | 68                                             |                       |          |       |
| Length Byte                               | 3                                                                                          | 2     | 2A                                             | 42                    |          |       |
| C Field                                   | 5                                                                                          | 2     | 44                                             |                       |          |       |
| M Field                                   | 7                                                                                          | 4     | 2448                                           |                       |          |       |
| A Field                                   | 11                                                                                         | 12    | 785634120107                                   |                       |          |       |
| CI Control                                | 23                                                                                         | 2     | 7A                                             |                       |          |       |
| Access Number                             | 25                                                                                         | 2     | 28                                             |                       |          |       |
| Status                                    | 27                                                                                         | 2     | 00                                             |                       |          |       |
| Config Word                               | 29                                                                                         | 4     | 0000                                           |                       |          |       |
| AES Verify                                | 33                                                                                         | 4     | 2F2F                                           |                       |          |       |
| DIF                                       | 37                                                                                         | 2     | 0F                                             |                       |          |       |
| VIF                                       | 39                                                                                         | 2     | 7F                                             |                       |          |       |
|                                           |                                                                                            |       |                                                | HEX                   | DEC      |       |
| Data                                      | 41                                                                                         | 46    | 11050B2010000200050001004020000003E8000003E801 | ID                    | 11       | 17    |
|                                           |                                                                                            |       |                                                | Device Type           | 05       | 5     |
|                                           |                                                                                            |       |                                                | Frame Counter         | 0B       | 11    |
|                                           |                                                                                            |       |                                                | TX ID                 | 20100002 |       |
|                                           |                                                                                            |       |                                                | Message Num           | 00       | 0     |
|                                           |                                                                                            |       |                                                | Tx Interval           | 05       | 5     |
|                                           |                                                                                            |       |                                                | Re-Try Count          | 00       | 0     |
|                                           |                                                                                            |       |                                                | Receiver ID           | 01004020 |       |
|                                           |                                                                                            |       |                                                | Initial Pulse Count 1 | 000003E8 | 1000  |
|                                           |                                                                                            |       |                                                | Initial Pulse Count 2 | 000003E8 | 1000  |
| RSSI                                      | 87                                                                                         | 2     | 3B                                             |                       | 3B       | 59    |
| Stop                                      | 89                                                                                         | 2     | 16                                             |                       |          | -29.5 |

Below are typical request and response packets for this message.

Received Packet (same as over the air)

\$68 \$15 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$02 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$10** \$05 \$29 \$16

Host will Respond with the following:

\$20 \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$05 \$0B \$20 \$10 \$00 \$02 \$00 \$05 \$00 \$01 \$00 \$40 \$20 \$00 \$00 \$03 \$E8 \$00 \$00 \$03 \$E8 \$01

**Note the response from the host starts from the CI field to the end of the data field and the first byte of the packet is the length field.**

### 4.11.2 TX\_ENERGY Modbus Slave Installation

| Message Number | Transmit Interval | Retry Count | Recvr Address D0 | Recvr Address D1 | Recvr Address D2 | Recvr Address D3 | Modbus Slave ID | UART Setting | First Register Address D0 | First Register Address D1 | No. Of Register to Read | Modbus Readout Interval |
|----------------|-------------------|-------------|------------------|------------------|------------------|------------------|-----------------|--------------|---------------------------|---------------------------|-------------------------|-------------------------|
|----------------|-------------------|-------------|------------------|------------------|------------------|------------------|-----------------|--------------|---------------------------|---------------------------|-------------------------|-------------------------|

## Enless Data Transmission Message Format

|                |                |                |      |      |      |      |      |      |      |      |      |      |
|----------------|----------------|----------------|------|------|------|------|------|------|------|------|------|------|
| 0x01 –<br>0x0A | 0x05 –<br>0xFF | 0x00 –<br>0x01 | 0x44 | 0x33 | 0x22 | 0x11 | 0x64 | 0x83 | 0x52 | 0xC5 | 0x02 | 0x05 |
|----------------|----------------|----------------|------|------|------|------|------|------|------|------|------|------|

- **Message Number** – it will be 0x01 to 0x0A for Energy meter Modbus slave device parameter message number. Number of message depending on slave device entered by user in field installation tool cvs file.
- **Transmit Interval** – It will be 0x05 to 0xFF is the periodic data transmission interval for a transmitter device. User can set interval time from minimum 5 to 255 minute.
- **Retry Count** – it will be 0 or 1, if user set retry count as 1, transmitter will send same message after sending periodic message. If it's not set, there will not be any retry message after sending periodic message from transmitter device.
- **Receiver Address (D0 – D3)** – it will be 4 bytes of data received from receiver which is consists of transmitter address. In given example 0x11223344.
- **Modbus Slave ID** – It will be unsigned one byte data of Modbus slave Address. In the given example is 0x64 which is 100 (decimal) address of slave device is.
- **UART setting** – It will be unsigned one byte encoded data of slave device communication baud rate. In the given example is 0x83 which is 19200-baud rate, None-parity, 2-Stopbit, 7 bit–Data bit, Parity Error Char-Ignore.

UART setting data decode example:

0x83 -> binary (MSB) 1 0 0 0 0 0 1 1 (LSB )  
D7 D6 D5 D4 D3 D2 D1 D0

LSB 3 bits (D2, D1, and D0) are consists of baud rate selection see below for more details.

D2 D1 D0  
 0 0 0 -> 2400 baud rate  
 0 0 1 -> 4800 baud rate  
 0 1 0 -> 9600 baud rate  
 0 1 1 -> 19200 baud rate  
 1 0 0 -> 38400 baud rate  
 1 0 1 -> Not Applicable  
 1 1 0 -> Not Applicable  
 1 1 1 -> Not Applicable

MSB 2 bits (D7, D6) are consists of stop bit selection see below for more details.

D7 D6  
 0 0 -> Stop bit 1  
 0 1 -> Not Applicable  
 1 0 -> Stop bit 2  
 1 1 -> Not Applicable

MSB 2 bits (D5, D4) are consists of Parity bit selection see below for more details.

D5 D4

0 0 -> Parity None

0 1 -> Parity Even

1 0 -> Parity Odd

1 1 -> Not Applicable

D3 Data bit

0 – 7 Data bits

1 – 8 Data bits

## Enless Data Transmission Message Format

A typical packet (over-the-air) breakdown is shown below:

| EMT MODBUS TRANSMITTER INSTALLATION PACKET |                                                                                   |       |                                       |                    |          |                                        |
|--------------------------------------------|-----------------------------------------------------------------------------------|-------|---------------------------------------|--------------------|----------|----------------------------------------|
| Data Frame                                 | 68264424487856341201077A280000002F2F0F7F11050B2010000203050001004020678A0B8053F16 |       |                                       |                    |          |                                        |
|                                            | Start                                                                             | Count | HEX VALUE                             | DEC                |          |                                        |
| Start Byte                                 | 1                                                                                 | 2     | 68                                    |                    |          |                                        |
| Length Byte                                | 3                                                                                 | 2     | 26                                    | 38                 |          |                                        |
| C Field                                    | 5                                                                                 | 2     | 44                                    |                    |          |                                        |
| M Field                                    | 7                                                                                 | 4     | 2448                                  |                    |          |                                        |
| A Field                                    | 11                                                                                | 12    | 785634120107                          |                    |          |                                        |
| CI Control                                 | 23                                                                                | 2     | 7A                                    |                    |          |                                        |
| Access Number                              | 25                                                                                | 2     | 28                                    |                    |          |                                        |
| Status                                     | 27                                                                                | 2     | 00                                    |                    |          |                                        |
| Config Word                                | 29                                                                                | 4     | 0000                                  |                    |          |                                        |
| AES Verify                                 | 33                                                                                | 4     | 2F2F                                  |                    |          |                                        |
| DIF                                        | 37                                                                                | 2     | 0F                                    |                    |          |                                        |
| VIF                                        | 39                                                                                | 2     | 7F                                    |                    |          |                                        |
|                                            |                                                                                   |       |                                       |                    | HEX      | DEC                                    |
| Data                                       | 41                                                                                | 38    | 11050B2010000203050001004020678A0B805 | ID                 | 11       | 17                                     |
|                                            |                                                                                   |       |                                       | Device Type        | 05       | 5                                      |
|                                            |                                                                                   |       |                                       | Frame Counter      | 0B       | 11                                     |
|                                            |                                                                                   |       |                                       | TX ID              | 20100002 | EMT Address                            |
|                                            |                                                                                   |       |                                       | Message Num        | 03       | 3                                      |
|                                            |                                                                                   |       |                                       | Tx Interval        | 05       | 5                                      |
|                                            |                                                                                   |       |                                       | Re-Try Count       | 00       | 0                                      |
|                                            |                                                                                   |       |                                       | Receiver ID        | 01004020 | Modbus Slave 3 Configuration           |
|                                            |                                                                                   |       |                                       | Mod Slave ID       | 67       | 103                                    |
|                                            |                                                                                   |       |                                       | UART Settings      | 8A       | 138                                    |
|                                            |                                                                                   |       |                                       | First Register Add | 0BB8     | 3000                                   |
|                                            |                                                                                   |       |                                       | Num Regs.          | 05       | 5                                      |
|                                            |                                                                                   |       |                                       |                    |          | Number of Registers to Read from Slave |
| RSSI                                       | 79                                                                                | 2     | 3F                                    |                    | 3F       | 63                                     |
| Stop                                       | 81                                                                                | 2     | 16                                    |                    |          | -31.5                                  |

Below are typical request and response packets for this message.

Received Packet (same as over the air) :

\$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$06 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$12 \$05 \$03 \$22 \$16

Host will Respond with the following:

\$1C \$7A \$74 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$11 \$05 \$0B \$20 \$10 \$00 \$02 \$04 \$05 \$00 \$01 \$00 \$40 \$20 \$68 \$8A \$0F \$A0 \$05

**Note the response from the host starts from the CI field to the end of the data field and the first byte of the packet is the length field.**

## 4.12 Installation ACK

| Message ID | Device Type | Frame Count |
|------------|-------------|-------------|
| 0x12       | 0x01 – 0xFF | 0x00-0xFF   |

- **Message ID** – It will be always 0x12 – Installation ACK
- **Device Type** – It will be 0x01 – 0xFF depending on transmitter type
- **Frame Count** – Increments to request for more data of next RSSI test depending on transmitter type

A typical packet (over-the-air) breakdown is shown below

| INSTALLATION ACKNOWLEDGEMENT PACKET |                                                    |              |                  |                    |            |            |       |
|-------------------------------------|----------------------------------------------------|--------------|------------------|--------------------|------------|------------|-------|
| <b>Data Frame</b>                   | 681644AE0C9900901001077A020000002F2F0F7F1204004C16 |              |                  |                    |            |            |       |
|                                     | <b>Start</b>                                       | <b>Count</b> | <b>HEX VALUE</b> | <b>DEC</b>         |            |            |       |
| <b>Start Byte</b>                   | 1                                                  | 2            | 68               |                    |            |            |       |
| <b>Length Byte</b>                  | 3                                                  | 2            | 16               | 22                 |            |            |       |
| <b>C Field</b>                      | 5                                                  | 2            | 44               |                    |            |            |       |
| <b>M Field</b>                      | 7                                                  | 4            | AE0C             |                    |            |            |       |
| <b>A Field</b>                      | 11                                                 | 12           | 990090100107     |                    |            |            |       |
| <b>CI Control</b>                   | 23                                                 | 2            | 7A               |                    |            |            |       |
| <b>Access Number</b>                | 25                                                 | 2            | 02               |                    |            |            |       |
| <b>Status</b>                       | 27                                                 | 2            | 00               |                    |            |            |       |
| <b>Config Word</b>                  | 29                                                 | 4            | 0000             |                    |            |            |       |
| <b>AES Verify</b>                   | 33                                                 | 4            | 2F2F             |                    |            |            |       |
| <b>DIF</b>                          | 37                                                 | 2            | 0F               |                    |            |            |       |
| <b>VIF</b>                          | 39                                                 | 2            | 7F               |                    |            |            |       |
|                                     |                                                    |              |                  |                    | <b>HEX</b> | <b>DEC</b> |       |
| <b>Data</b>                         | 41                                                 | 6            | 120400           | <b>ID</b>          | 12         | 18         |       |
|                                     |                                                    |              |                  | <b>Device Type</b> | 04         | 4          |       |
|                                     |                                                    |              |                  | <b>Frame Count</b> | 00         | 0          |       |
|                                     |                                                    |              |                  |                    |            |            |       |
| <b>RSSI</b>                         | 47                                                 | 2            | 4C               |                    | 4C         | 76         | -38.0 |
| <b>Stop</b>                         | 49                                                 | 2            | 16               |                    |            |            |       |

This message is received by the host and same as 'over-the-air message':

\$68 \$16 \$44 \$AE \$0C \$02 \$00 \$10 \$20 \$01 \$07 \$7A \$06 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$12 \$05 \$03 \$22 \$16

### 4.13 Installation RSSI Request

| Message ID | Device Type | Frame Count |
|------------|-------------|-------------|
| 0x14       | 0x01 – 0xFF | 0x00-0xFF   |

- **Message ID** – It will be always 0x14 – RSSI Request
- **Device Type** – It will be 0x01 – 0xFF depending on transmitter type
- **Frame Count** – Increments to request for more data of next RSSI test depending on transmitter type

A typical packet (over-the-air) breakdown is shown below:

| INSTALLATION RSSI REQUEST PACKET |                                                    |       |              |             |     |     |       |
|----------------------------------|----------------------------------------------------|-------|--------------|-------------|-----|-----|-------|
| Data Frame                       | 681644AE0C9900901001077A030000002F2F0F7F1404025416 |       |              |             |     |     |       |
|                                  | Start                                              | Count | HEX VALUE    | DEC         |     |     |       |
| Start Byte                       | 1                                                  | 2     | 68           |             |     |     |       |
| Length Byte                      | 3                                                  | 2     | 16           | 22          |     |     |       |
| C Field                          | 5                                                  | 2     | 44           |             |     |     |       |
| M Field                          | 7                                                  | 4     | AE0C         |             |     |     |       |
| A Field                          | 11                                                 | 12    | 990090100107 |             |     |     |       |
| CI Control                       | 23                                                 | 2     | 7A           |             |     |     |       |
| Access Number                    | 25                                                 | 2     | 03           |             |     |     |       |
| Status                           | 27                                                 | 2     | 00           |             |     |     |       |
| Config Word                      | 29                                                 | 4     | 0000         |             |     |     |       |
| AES Verify                       | 33                                                 | 4     | 2F2F         |             |     |     |       |
| DIF                              | 37                                                 | 2     | 0F           |             |     |     |       |
| VIF                              | 39                                                 | 2     | 7F           |             |     |     |       |
|                                  |                                                    |       |              |             | HEX | DEC |       |
| Data                             | 41                                                 | 6     | 140402       | ID          | 14  | 20  |       |
|                                  |                                                    |       |              | Device Type | 04  | 4   |       |
|                                  |                                                    |       |              | Frame Count | 02  | 2   |       |
|                                  |                                                    |       |              |             |     |     |       |
| RSSI                             | 47                                                 | 2     | 54           |             | 54  | 84  | -42.0 |
| Stop                             | 49                                                 | 2     | 16           |             |     |     |       |

This message is received by the host and same as 'over-the-air message':

\$68 \$16 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$06 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$14 \$03 \$05 \$1F \$16



#### 4.14 RSSI Response

| Message ID | Device Type | Frame Count | Transmitter ID | RSSI RX |
|------------|-------------|-------------|----------------|---------|
| 0x15       | 0x01 – 0xFF | 0x00-0xFF   | XXXXXXXX       | 0x3A    |

- **Message ID** – It will be always 0x15 – RSSI Response
- **Device Type** – It will be 0x01 – 0xFF depending on transmitter type
- **Frame Count** – Increments to request for more data of next RSSI test depending on transmitter type
- **Transmitter ID** – ID of the transmitter that is being Installed
- **RSSI RX** – This is the received RSSI level from the transmitter. It is sent back to the transmitter as validation of the signal level between Transmitter and Receiver.

A typical packet (over-the-air) breakdown is shown below:

| RSSI RESPONSE PACKET |                                                              |              |                  |                      |            |            |          |
|----------------------|--------------------------------------------------------------|--------------|------------------|----------------------|------------|------------|----------|
| <b>Data Frame</b>    | 681B4424487856341201077 000000002F2F0F7F150401109000993A3916 |              |                  |                      |            |            |          |
|                      | <b>Start</b>                                                 | <b>Count</b> | <b>HEX VALUE</b> | <b>DEC</b>           |            |            |          |
| <b>Start Byte</b>    | 1                                                            | 2            | 68               |                      |            |            |          |
| <b>Length Byte</b>   | 3                                                            | 2            | 1B               | 27                   |            |            |          |
| <b>C Field</b>       | 5                                                            | 2            | 44               |                      |            |            |          |
| <b>M Field</b>       | 7                                                            | 4            | 2448             |                      |            |            |          |
| <b>A Field</b>       | 11                                                           | 12           | 785634120107     |                      |            |            |          |
| <b>CI Control</b>    | 23                                                           | 2            | 7                |                      |            |            |          |
| <b>Access Number</b> | 25                                                           | 2            | 00               |                      |            |            |          |
| <b>Status</b>        | 27                                                           | 2            | 00               |                      |            |            |          |
| <b>Config Word</b>   | 29                                                           | 4            | 0000             |                      |            |            |          |
| <b>AES Verify</b>    | 33                                                           | 4            | 2F2F             |                      |            |            |          |
| <b>DIF</b>           | 37                                                           | 2            | 0F               |                      |            |            |          |
| <b>VIF</b>           | 39                                                           | 2            | 7F               |                      |            |            |          |
|                      |                                                              |              |                  |                      | <b>HEX</b> | <b>DEC</b> |          |
| <b>Data</b>          | 41                                                           | 16           | 150401109000993A | <b>ID</b>            | 15         | 21         |          |
|                      |                                                              |              |                  | <b>Device Type</b>   | 04         | 4          |          |
|                      |                                                              |              |                  | <b>Frame Counter</b> | 01         | 1          |          |
|                      |                                                              |              |                  | <b>TX ID</b>         | 10900099   |            | 10900099 |
|                      |                                                              |              |                  | <b>RSSI Rx'd</b>     | 3A         | 58         |          |
| <b>RSSI</b>          | 57                                                           | 2            | 39               |                      | 39         | 57         | -28.5    |
| <b>Stop</b>          | 59                                                           | 2            | 16               |                      |            |            |          |

## Enless Data Transmission Message Format

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Below are typical request and response packets for this message.

Received Packet (same as over the air) :

\$68 \$16 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$06 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$14** \$03 \$05 \$1F \$16

Host will Respond with the following:

\$11 \$7A \$62 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$15** \$03 \$05 \$10 \$50 \$00 \$18 \$26

**Note the response from the host starts from the CI field to the end of the data field and the first byte of the packet is the length field.**

### 4.15 Installation Successful

| Message ID | Device Type | Frame Count |
|------------|-------------|-------------|
| 0x14       | 0x01 – 0xFF | 0x00-0xFF   |

- **Message ID** – It will be always 0x16 – Installation Successful
- **Device Type** – It will be 0x01 – 0xFF depending on transmitter type
- **Frame Count** – Increments to request for more data of next RSSI test depending on transmitter type

A typical packet (over-the-air) breakdown is shown below:

| INSTALLATION SUCCESSFUL MESSAGE |                                                  |              |                  |                    |            |            |       |
|---------------------------------|--------------------------------------------------|--------------|------------------|--------------------|------------|------------|-------|
| <b>Data Frame</b>               | 681544AE0C9900901001077A070000002F2F0F7F16044016 |              |                  |                    |            |            |       |
|                                 | <b>Start</b>                                     | <b>Count</b> | <b>HEX VALUE</b> | <b>DEC</b>         |            |            |       |
| <b>Start Byte</b>               | 1                                                | 2            | 68               |                    |            |            |       |
| <b>Length Byte</b>              | 3                                                | 2            | 15               | 21                 |            |            |       |
| <b>C Field</b>                  | 5                                                | 2            | 44               |                    |            |            |       |
| <b>M Field</b>                  | 7                                                | 4            | AE0C             |                    |            |            |       |
| <b>A Field</b>                  | 11                                               | 12           | 990090100107     |                    |            |            |       |
| <b>CI Control</b>               | 23                                               | 2            | 7A               |                    |            |            |       |
| <b>Access Number</b>            | 25                                               | 2            | 07               |                    |            |            |       |
| <b>Status</b>                   | 27                                               | 2            | 00               |                    |            |            |       |
| <b>Config Word</b>              | 29                                               | 4            | 0000             |                    |            |            |       |
| <b>AES Verify</b>               | 33                                               | 4            | 2F2F             |                    |            |            |       |
| <b>DIF</b>                      | 37                                               | 2            | 0F               |                    |            |            |       |
| <b>VIF</b>                      | 39                                               | 2            | 7F               |                    |            |            |       |
|                                 |                                                  |              |                  |                    | <b>HEX</b> | <b>DEC</b> |       |
| <b>Data</b>                     | 41                                               | 4            | 1604             | <b>ID</b>          | 16         | 22         |       |
|                                 |                                                  |              |                  | <b>Device Type</b> | 04         | 4          |       |
|                                 |                                                  |              |                  |                    |            |            |       |
| <b>RSSI</b>                     | 45                                               | 2            | 40               |                    | 40         | 64         | -32.0 |
| <b>Stop</b>                     | 47                                               | 2            | 16               |                    |            |            |       |

This message is received by the host and same as 'over-the-air message':

\$68 \$15 \$44 \$AE \$0C \$18 \$00 \$50 \$10 \$01 \$07 \$7A \$07 \$00 \$00 \$00 \$2F \$2F \$0F \$7F \$16 \$03 \$1F \$16

#### 4.16 RX\_MODBUS Modbus Receiver Installation

| Modbus Device Address | Modbus Register Table 1 Start Address High Byte | Modbus Register Table 1 Start Address Low Byte | Energy Meter Register Table2 Start Address High byte | Energy Meter Register Table2 Start Address Low byte | Installed Device ID (D0 – D3) | Energy Meter Slave ID | Installed Device ID (D0 – D3) | Energy Meter Slave ID | ... | Installed Device ID (D0 – D3) | Energy Meter Slave ID |
|-----------------------|-------------------------------------------------|------------------------------------------------|------------------------------------------------------|-----------------------------------------------------|-------------------------------|-----------------------|-------------------------------|-----------------------|-----|-------------------------------|-----------------------|
| 0x15                  | 0x03                                            | 0xE8                                           | 0x07                                                 | 0xD0                                                | 0x11223344                    | 0xFF                  | 0x55667788                    | 0x64                  | ... | 0x99001122                    | 0xFF                  |
|                       |                                                 |                                                |                                                      |                                                     | 1                             | 1                     | 2                             | 2                     | ... | 32                            | 32                    |

- **Modbus Device Address** – it will be unsigned byte of data which is Receiver Modbus Device address for polling data. In the given example 0x15 which is 21(decimal).
- **Modbus Register Table 1 Start Address low and high byte** – It will be unsigned word data to indicate modbus register table 1 start address. In given example is 0x03E8 which is 1000(decimal). In this table 1 contains all transmitter data except energy meter slave data.
- **Energy Meter Register Table 2 Address low and high byte** – It will be unsigned word data to indicate modbus register table 2 start address. In the given example 0x07D0 which is 2000(decimal) . In this table 2 contains all energy meter slave device data.
- **Installed Device ID** – It will be installed device id of 4 byte data of all transmitter devices in CSV file, including energy meter slave device. In the given example 0x11223344.
- **Energy Meter Slave ID** – It will be unsigned one byte data of Energy meter slave Id, it will be always 0xFF if there is no energy meter slave device. In the given example 0xFF is general transmitter device and 0x64 which is 100 (decimal) energy meter slave device id.
- **Installed Device Id and Energy meter Slave ID will repeat depending on no of transmitter device installed in CSV file except Receiver ID and its limited to 1 to 32 transmitter devices as shown in above table.**

## Enless Data Transmission Message Format

Below is a typical sequence seen on the field installation tool for all transmitters, except the Energy Meter Transmitter:

```
26/05/2014 09:13.23 <INSTALLREQ> Installation request message received from 20400001
26/05/2014 09:13.23 <INSTALLPARAMS> Installation Parameters sent to 20400001
26/05/2014 09:13.26 <INSTALSUCC> Installation is successful for 20400001
26/05/2014 09:13.55 <INSTALLREQ> Installation request message received from 20400001
26/05/2014 09:13.55 <INSTALLPARAMS> Installation Parameters sent to 20400001
26/05/2014 09:13.58 <INSTALSUCC> Installation is successful for 20400001
26/05/2014 09:14.27 <INSTALLREQ> Installation request message received from 20400001
26/05/2014 09:14.27 <INSTALLPARAMS> Installation Parameters sent to 20400001
26/05/2014 09:14.30 <INSTALSUCC> Installation is successful for 20400001
```

The sequence of communications is as follows:

| No. | Time     | Direction | Description                            | Message Type                   |
|-----|----------|-----------|----------------------------------------|--------------------------------|
| 1   | 09:13:23 | Tx→Host   | Installation Request Received from Rx. | Installation Request (0x10)    |
| 2   | 09:13:23 | Host→Tx   | Installation Parameters Sent to Tx.    | Installation Parameter (0x11)  |
| 3   |          | Tx→Host   | Installation Request Received from Rx. | Installation Request (0x10)    |
| 4   |          | Host→Tx   | Installation Parameters Sent to Tx.    | Installation Parameter (0x11)  |
| 5   | 12:13:26 | Tx→Host   | Installation Successful                | Installation Successful (0x16) |
| 6   |          | Tx→Host   | Installation Successful                | Installation Successful (0x16) |

The parameters sent to the Modbus Receiver will depend on the configuration required.

As an example, below is the CSV file used:

| Device ID | Device type   | Periodicity | Retry count | TX PULSE:<br>Initial count<br>1 | TX PULSE:<br>Initial count<br>2 | TX ENERGY:<br>Slave ID | Baudrate | Parity | Stopbits | Databits | TX ENERGY:<br>Register<br>Nbr | TX ENERGY<br>: Register<br>Cnt | RX<br>MODBUS ID | RX<br>MODBUS:<br>First<br>register | RX<br>MODBUS:<br>First TX<br>ENERGY<br>Register |
|-----------|---------------|-------------|-------------|---------------------------------|---------------------------------|------------------------|----------|--------|----------|----------|-------------------------------|--------------------------------|-----------------|------------------------------------|-------------------------------------------------|
| 10100013  | TX TEMP       | 5           | 0           | NA                              | NA                              | NA                     | NA       | NA     | NA       | NA       | NA                            | NA                             | NA              | NA                                 | NA                                              |
| 10100578  | TX TEMP       | 5           | 0           | NA                              | NA                              | NA                     | NA       | NA     | NA       | NA       | NA                            | NA                             | NA              | NA                                 | NA                                              |
| 10300023  | TX TEMP       | 5           | 0           | NA                              | NA                              | NA                     | NA       | NA     | NA       | NA       | NA                            | NA                             | NA              | NA                                 | NA                                              |
| 10200061  | TX TEMP HUM   | 5           | 0           | NA                              | NA                              | NA                     | NA       | NA     | NA       | NA       | NA                            | NA                             | NA              | NA                                 | NA                                              |
| 10200067  | TX TEMP HUM   | 5           | 0           | NA                              | NA                              | NA                     | NA       | NA     | NA       | NA       | NA                            | NA                             | NA              | NA                                 | NA                                              |
| 10500086  | TX TEMP PT100 | 5           | 0           | NA                              | NA                              | NA                     | NA       | NA     | NA       | NA       | NA                            | NA                             | NA              | NA                                 | NA                                              |
| 10500028  | TX TEMP PT100 | 5           | 0           | NA                              | NA                              | NA                     | NA       | NA     | NA       | NA       | NA                            | NA                             | NA              | NA                                 | NA                                              |
| 10500018  | TX TEMP PT100 | 5           | 0           | NA                              | NA                              | NA                     | NA       | NA     | NA       | NA       | NA                            | NA                             | NA              | NA                                 | NA                                              |
| 10000051  | TX PULSE      | 5           | 0           | 0                               | 0                               | NA                     | NA       | NA     | NA       | NA       | NA                            | NA                             | NA              | NA                                 | NA                                              |
| 10900099  | TX PULSE      | 5           | 0           | 10                              | 100                             | NA                     | NA       | NA     | NA       | NA       | NA                            | NA                             | NA              | NA                                 | NA                                              |
| 10900001  | TX PULSE      | 5           | 0           | 0                               | 0                               | NA                     | NA       | NA     | NA       | NA       | NA                            | NA                             | NA              | NA                                 | NA                                              |
| 20100002  | TX ENERGY     | 5           | 0           | 1000                            | 1000                            | NA                     | NA       | NA     | NA       | NA       | NA                            | NA                             | 1               | NA                                 | NA                                              |
| 20100002  | TX ENERGY     | 5           | 0           | NA                              | NA                              | 101                    | 9600     | None   | 2        | 8        | 1000                          | 5                              | NA              | NA                                 | NA                                              |
| 20100002  | TX ENERGY     | 5           | 0           | NA                              | NA                              | 102                    | 9600     | None   | 2        | 8        | 2000                          | 5                              | NA              | NA                                 | NA                                              |
| 20100002  | TX ENERGY     | 5           | 0           | NA                              | NA                              | 103                    | 9600     | None   | 2        | 8        | 3000                          | 5                              | NA              | NA                                 | NA                                              |
| 20100002  | TX ENERGY     | 5           | 0           | NA                              | NA                              | 104                    | 9600     | None   | 2        | 8        | 4000                          | 5                              | NA              | NA                                 | NA                                              |
| 20100002  | TX ENERGY     | 5           | 0           | NA                              | NA                              | 105                    | 9600     | None   | 2        | 8        | 5000                          | 5                              | NA              | NA                                 | NA                                              |
| 20100002  | TX ENERGY     | 5           | 0           | NA                              | NA                              | 106                    | 9600     | None   | 2        | 8        | 6000                          | 5                              | NA              | NA                                 | NA                                              |
| 20100002  | TX ENERGY     | 5           | 0           | NA                              | NA                              | 107                    | 9600     | None   | 2        | 8        | 7000                          | 5                              | NA              | NA                                 | NA                                              |
| 20100002  | TX ENERGY     | 5           | 0           | NA                              | NA                              | 108                    | 9600     | None   | 2        | 8        | 8000                          | 5                              | NA              | NA                                 | NA                                              |
| 20100002  | TX ENERGY     | 5           | 0           | NA                              | NA                              | 109                    | 9600     | None   | 2        | 8        | 9000                          | 5                              | NA              | NA                                 | NA                                              |
| 20100002  | TX ENERGY     | 5           | 0           | NA                              | NA                              | 110                    | 9600     | None   | 2        | 8        | 10000                         | 5                              | NA              | NA                                 | NA                                              |
| 20400001  | RX MODBUS     | NA          | NA          | NA                              | NA                              | NA                     | 9600     | None   | 2        | 8        | NA                            | NA                             | 1               | 31000                              | 41000                                           |

Some Complete Installation Sequence Packets (with the Enless Receiver) can be seen below (with the packet types highlighted in Red:

RX: Messages Received by Host from Enless Receiver

TX: Messages sent Out from Host to Enless Receiver

**RX:** \$68 \$15 \$20 \$AE \$0C \$01 \$00 \$40 \$20 \$0C \$AE \$7A \$02 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$10** \$07 \$FF \$16

**TX:** \$68 \$89 \$FF \$AE \$0C \$FF \$FF \$FF \$FF \$0C \$AE \$7A \$68 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$07 \$8A \$01 \$18 \$79 \$28 \$A0 \$13 \$00 \$10 \$10 \$FF \$78 \$05 \$10 \$10 \$FF \$23 \$00 \$30 \$10 \$FF \$61 \$00 \$20 \$10 \$FF \$67 \$00 \$20 \$10 \$FF \$86 \$00 \$50 \$10 \$FF \$28 \$00 \$50 \$10 \$FF \$18 \$00 \$50 \$10 \$FF \$51 \$00 \$00 \$10 \$FF \$99 \$00 \$90 \$10 \$FF \$01 \$00 \$90 \$10 \$FF \$02 \$00 \$10 \$20 \$FF \$02 \$00 \$10 \$20 \$65 \$02 \$00 \$10 \$20 \$66 \$02 \$00 \$10 \$20 \$67 \$02 \$00 \$10 \$20 \$68 \$02 \$00 \$10 \$20 \$69 \$02 \$00 \$10 \$20 \$6A \$02 \$00 \$10 \$20 \$6B \$02 \$00 \$10 \$20 \$6C \$02 \$00 \$10 \$20 \$6D \$02 \$00 \$10 \$20 \$6E \$FF \$16

**RX:** \$68 \$15 \$20 \$AE \$0C \$01 \$00 \$40 \$20 \$0C \$AE \$7A \$03 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$10** \$07 \$FF \$16

**TX:** \$68 \$89 \$FF \$AE \$0C \$FF \$FF \$FF \$FF \$0C \$AE \$7A \$68 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$11** \$07 \$8A \$01 \$18 \$79 \$28 \$A0 \$13 \$00 \$10 \$10 \$FF \$78 \$05 \$10 \$10 \$FF \$23 \$00 \$30 \$10 \$FF \$61 \$00 \$20 \$10 \$FF \$67 \$00 \$20 \$10 \$FF \$86 \$00 \$50 \$10 \$FF \$28 \$00 \$50 \$10 \$FF \$18 \$00 \$50 \$10 \$FF \$51 \$00 \$00 \$10 \$FF \$99 \$00 \$90 \$10 \$FF \$01 \$00 \$90 \$10 \$FF \$02 \$00 \$10 \$20 \$FF \$02 \$00 \$10 \$20 \$65 \$02 \$00 \$10 \$20 \$66 \$02 \$00 \$10 \$20 \$67 \$02 \$00 \$10 \$20 \$68 \$02 \$00 \$10 \$20 \$69 \$02 \$00 \$10 \$20 \$6A \$02 \$00 \$10 \$20 \$6B \$02 \$00 \$10 \$20 \$6C \$02 \$00 \$10 \$20 \$6D \$02 \$00 \$10 \$20 \$6E \$FF \$16

**RX:** \$68 \$15 \$20 \$AE \$0C \$01 \$00 \$40 \$20 \$0C \$AE \$7A \$04 \$00 \$00 \$00 \$2F \$2F \$0F \$7F **\$16** \$07 \$FF \$16

## Enless Data Transmission Message Format

The message from the Receiver to host complies with the same format as the transmitters

The message from the host to the receiver also complies to the full WMBUS format as shown below:

| MODBUS RECEIVER INSTALLATION PACKET |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  |                       |       |       |            |          |                                |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------|-------|------------|----------|--------------------------------|
| Data Frame                          | 6889FFAEOCFFFFFFFFFOCAE7A68000002F2F0F7F11078A01187928A013001010FF78051010FF23003010FF61002010FF67002010FF86005010FF28005010FF18005010FF51000010FF99009010FF01009010FF02001020650200102066020010206702001020680200102069020010206A020010206B020010206C020010206D020010206E020010206EFF16 |       |                                                                                                                                                                                                                                                  |                       |       |       |            |          |                                |
|                                     | Start                                                                                                                                                                                                                                                                                    | Count | HEX VALUE                                                                                                                                                                                                                                        | DEC                   |       |       |            |          |                                |
| Start Byte                          | 1                                                                                                                                                                                                                                                                                        | 2     | 68                                                                                                                                                                                                                                               |                       |       |       |            |          |                                |
| Length Byte                         | 3                                                                                                                                                                                                                                                                                        | 2     | 89                                                                                                                                                                                                                                               | 137                   |       |       |            |          |                                |
| C Field                             | 5                                                                                                                                                                                                                                                                                        | 2     | FF                                                                                                                                                                                                                                               |                       |       |       |            |          |                                |
| M Field                             | 7                                                                                                                                                                                                                                                                                        | 4     | AEOC                                                                                                                                                                                                                                             |                       |       |       |            |          |                                |
| A Field                             | 11                                                                                                                                                                                                                                                                                       | 12    | FFFFFFFFFOCAE                                                                                                                                                                                                                                    |                       |       |       |            |          |                                |
| CI Control                          | 23                                                                                                                                                                                                                                                                                       | 2     | 7A                                                                                                                                                                                                                                               |                       |       |       |            |          |                                |
| Access Number                       | 25                                                                                                                                                                                                                                                                                       | 2     | 68                                                                                                                                                                                                                                               |                       |       |       |            |          |                                |
| Status                              | 27                                                                                                                                                                                                                                                                                       | 2     | 00                                                                                                                                                                                                                                               |                       |       |       |            |          |                                |
| Config Word                         | 29                                                                                                                                                                                                                                                                                       | 4     | 0000                                                                                                                                                                                                                                             |                       |       |       |            |          |                                |
| AES Verify                          | 33                                                                                                                                                                                                                                                                                       | 4     | 2F2F                                                                                                                                                                                                                                             |                       |       |       |            |          |                                |
| DIF                                 | 37                                                                                                                                                                                                                                                                                       | 2     | 0F                                                                                                                                                                                                                                               |                       |       |       |            |          |                                |
| VIF                                 | 39                                                                                                                                                                                                                                                                                       | 2     | 7F                                                                                                                                                                                                                                               |                       |       |       |            |          |                                |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  |                       | Start | Count | HEX        | DEC      |                                |
| Data                                | 41                                                                                                                                                                                                                                                                                       | 236   | 11078A01187928A013001010FF78051010FF23003010FF61002010FF67002010FF86005010FF28005010FF18005010FF51000010FF99009010FF01009010FF02001020650200102066020010206702001020680200102069020010206A020010206B020010206C020010206D020010206E020010206EFF16 | ID                    | 41    | 2     | 11         | 17       |                                |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Device Type           | 43    | 2     | 07         | 7        |                                |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | UART Settings         | 45    | 2     | 8A         | 138      |                                |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Modbus ID             | 47    | 2     | 01         |          | Modbus receiver Modbus ID      |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Table 1 Start         | 49    | 4     | 1879       | 31000    | Modbus Table 1 Start           |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Table 2 Start         | 53    | 4     | 28A0       | 41000    | Modbus Table 2 Start           |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 57    | 10    | 13001010FF | 10100013 | Type FF - Standard Transmitter |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 67    | 10    | 78051010FF | 10100578 | Type FF - Standard Transmitter |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 77    | 10    | 23003010FF | 10300023 | Type FF - Standard Transmitter |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 87    | 10    | 61002010FF | 10200061 | Type FF - Standard Transmitter |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 97    | 10    | 67002010FF | 10200067 | Type FF - Standard Transmitter |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 107   | 10    | 86005010FF | 10500086 | Type FF - Standard Transmitter |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 117   | 10    | 28005010FF | 10500028 | Type FF - Standard Transmitter |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 127   | 10    | 18005010FF | 10500018 | Type FF - Standard Transmitter |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 137   | 10    | 51000010FF | 10000051 | Type FF - Standard Transmitter |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 147   | 10    | 99009010FF | 10900099 | Type FF - Standard Transmitter |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 157   | 10    | 01009010FF | 10900001 | Type FF - Standard Transmitter |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 167   | 10    | 02001020FF | 20100002 | Type FF - Standard Transmitter |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 177   | 10    | 0200102065 | 20100002 | EMT Slave Address 0x65 = 101   |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 187   | 10    | 0200102066 | 20100002 | EMT Slave Address 0x66 = 102   |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 197   | 10    | 0200102067 | 20100002 | EMT Slave Address 0x67 = 103   |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 207   | 10    | 0200102068 | 20100002 | EMT Slave Address 0x68 = 104   |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 217   | 10    | 0200102069 | 20100002 | EMT Slave Address 0x69 = 105   |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 227   | 10    | 020010206A | 20100002 | EMT Slave Address 0x6A = 106   |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 237   | 10    | 020010206B | 20100002 | EMT Slave Address 0x6B = 107   |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 247   | 10    | 020010206C | 20100002 | EMT Slave Address 0x6C = 108   |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 257   | 10    | 020010206D | 20100002 | EMT Slave Address 0x6D = 109   |
|                                     |                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                  | Transmitter ID / Type | 267   | 10    | 020010206E | 20100002 | EMT Slave Address 0x6E = 110   |
| RSSI                                | 277                                                                                                                                                                                                                                                                                      | 2     | FF                                                                                                                                                                                                                                               |                       |       |       | FF         | 255      | -127.5                         |
| Stop                                | 279                                                                                                                                                                                                                                                                                      | 2     | 16                                                                                                                                                                                                                                               |                       |       |       |            |          |                                |